



# **Video-Based Cryptanalysis**

**Extracting Cryptographic Keys from Video  
Footage of a Device's Power LED**

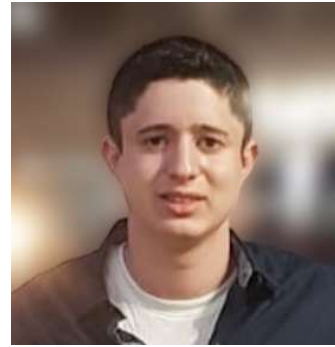
**Ben Nassi and Ofek Vayner**

# About Us



Ben Nassi

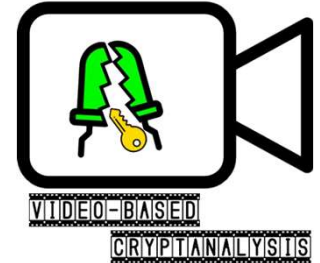
- ❖ Black Hat board member (Europe & Singapore)
- ❖ 4<sup>th</sup> Black Hat talk
- ❖ Freelancer consultant
- ❖ Postdoc @ Cornell Tech
- ❖ Ph.D. in security and privacy



Ofek Vayner

- ❖ M.Sc. student @ Ben-Gurion University of the Negev (BGU)

# About This Talk



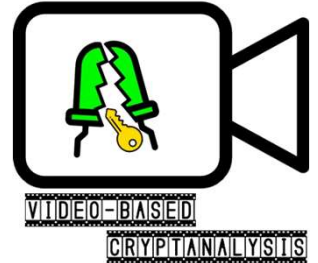
This talk introduces two recently published papers:

❖ ***Video-Based Cryptanalysis: Extracting Cryptographic***

***Keys from Video Footage of a Device's Power LED*** by: Ben Nassi, Etay Iluz, Or Cohen, Ofek Vayner, Dudi Nassi, Boris Zadov, Yuval Elovici

❖ ***Optical Cryptanalysis: Recovering Cryptographic Keys from Power LED Light Fluctuations*** (to be presented at CCS 2023) by: Ben Nassi, Ofek Vayner, Etay Iluz, Dudi Nassi, Or Hai Cohen, Jan Jancar, Daniel Genkin, Eran Tromer, Boris Zadov, Yuval Elovici

# About This Talk



1. No prior knowledge of cryptography is required to understand this talk.
2. Some details about the attack implementations aren't covered in this talk in order to keep it as simple as possible (you can find them in the paper).

With that in mind, let's have some fun!

# Introduction

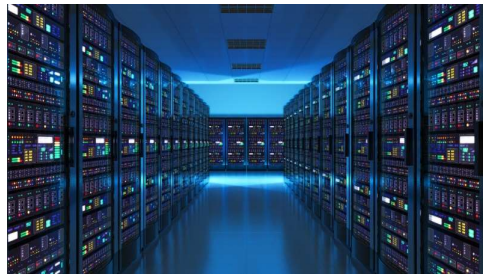
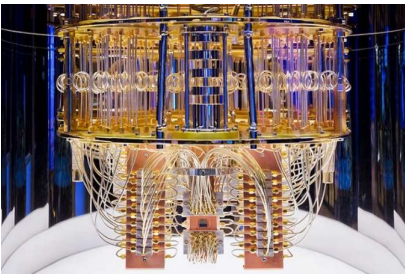


What do you associate with the term “cryptanalysis”?

# Introduction



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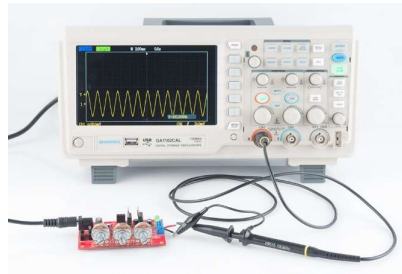
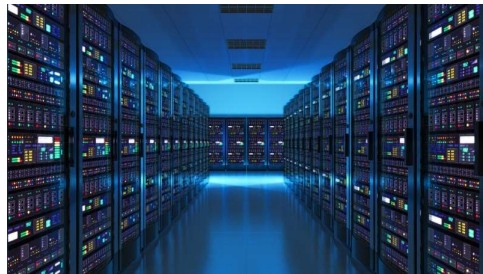
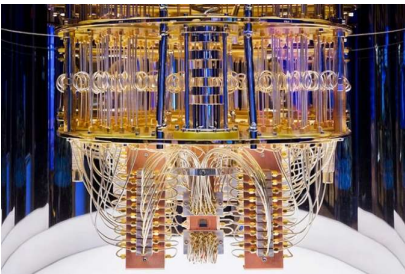


High level of  
Computing Capabilities

# Introduction



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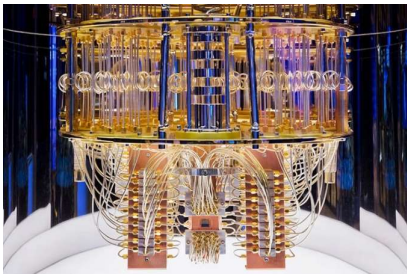
High level of  
Computing Capabilities

Specialized  
Hardware

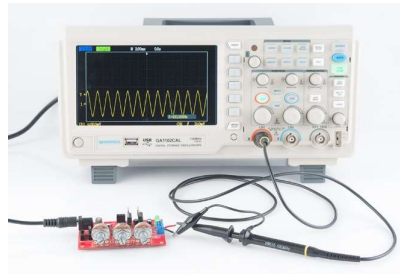
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High level of  
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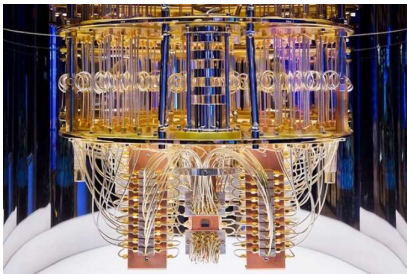
Complex  
Attacks



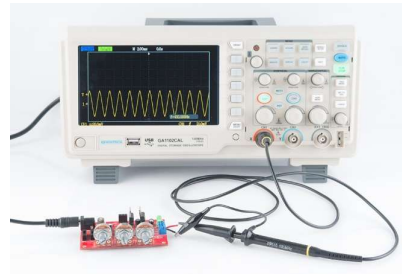
# Introduction



What do you associate with the term “cryptanalysis”?



High level of  
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Specialized  
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Complex  
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Special  
Agencies

# Introduction

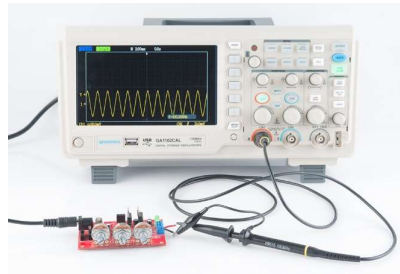
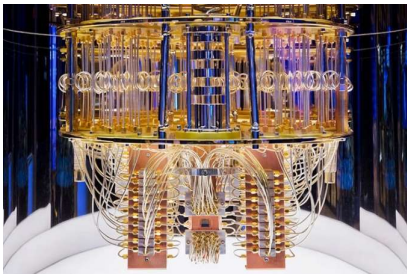


Weak  
Computing  
Capabilities

Popular  
Device

Cannot be used to  
apply complex  
attacks

Me???



High level of  
Computing Capabilities

Specialized  
Hardware

Complex  
Attacks

Special  
Agencies

# Introduction



Our message:  
think again



# Introduction



Power LEDs pose a risk to  
information confidentiality



Infrastructure to  
exploit

# Introduction



Where did the idea of using a video camera to recover secret keys from a device's power LED come from?

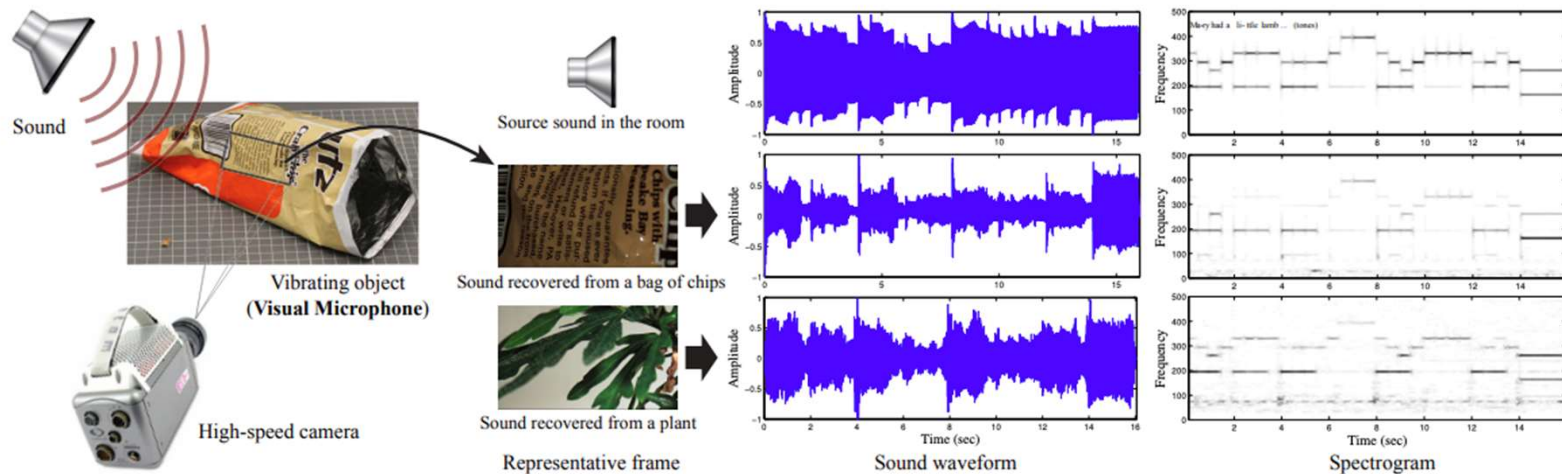
Power LEDs pose a risk to information confidentiality

Infrastructure to exploit

# Related Work

## ❖ 2014 - Visual Microphone (Davis et al.)

- ❖ Speech recovery using a video camera
- ❖ Analyzing an object's (e.g., a bag of chips) movement in response to speech in a video
- ❖ High-frequency video camera (20K FPS)



# Related Work

- ❖ 2014 - Visual Microphone (Davis et al.)
- ❖ 2020 - Lamphone (Nassi et al.) at Black Hat USA 2020
  - ❖ Speech recovery using a photodiode (optical sensor)
  - ❖ Analyzing a desktop light bulb's movement in response to speech
  - ❖ Speech was recovered from 35 meters



The photodiode  
used



"We will make America great again!"  
Original



# Related Work

- ❖ 2014 - Visual Microphone (Davis et al.)
- ❖ 2020 - Lamphone (Nassi et al.) at Black Hat USA 2020
- ❖ 2021 - Glowworm Attack (Nassi et al.) at CCS 2021
  - ❖ Speech recovery using a photodiode
  - ❖ Analyzing a speaker's power LED intensity when speech is played
  - ❖ Speech recovered from 35 meters



The photodiode  
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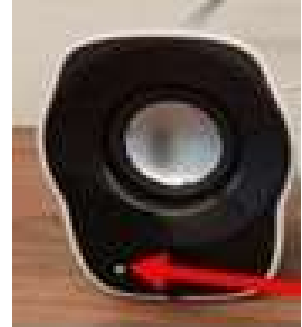




# Related Work

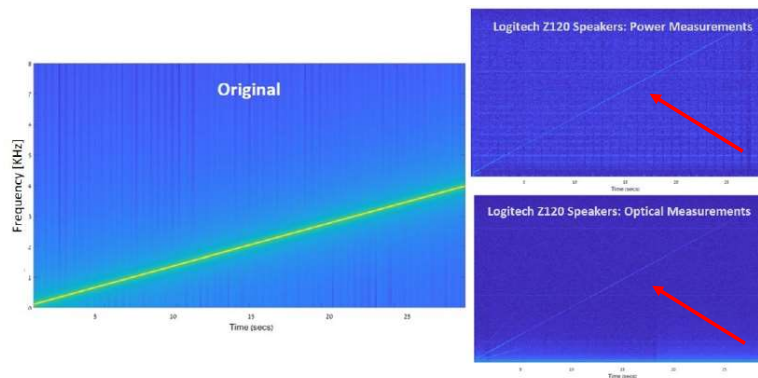
- ❖ 2021 - Glowworm Attack (Nassi et al.) at CCS 2021

Why can the intensity of a speaker's power LED be used to recover speech?



- ❖ Experiment

- ❖ USB speakers were used to play a frequency scan (0-4 kHz)
- ❖ Power traces were obtained from an adapter that was connected to the speaker's USB
- ❖ A photodiode was used to obtain an optical trace from the speaker's power LED



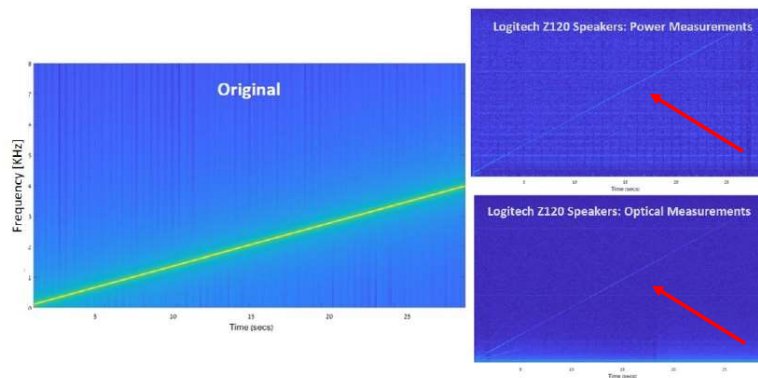
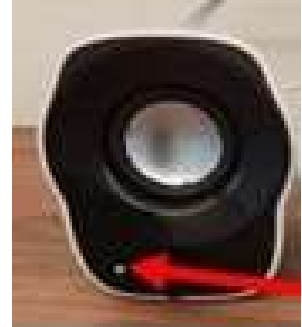
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The power consumption correlates with the intensity of the power LED.

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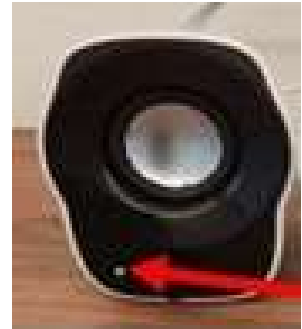
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Why can the intensity of a speaker's power LED be used to recover speech?

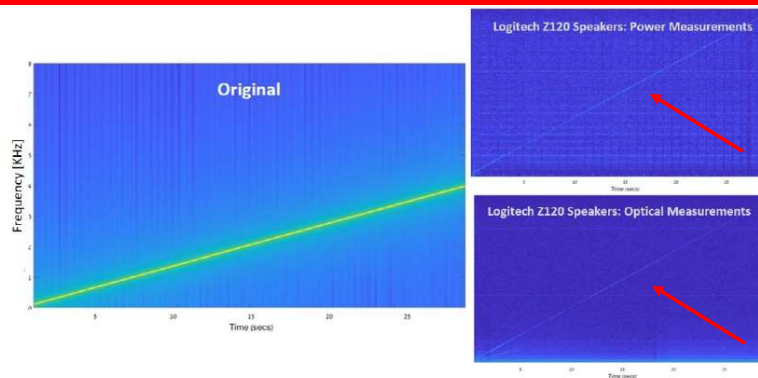
The power consumption correlates with the intensity of the power LED.

✗ USB speakers were used to play a frequency scan (0-4 kHz)

Is a correlation between the intensity of the power LED and power consumption seen in other speakers?



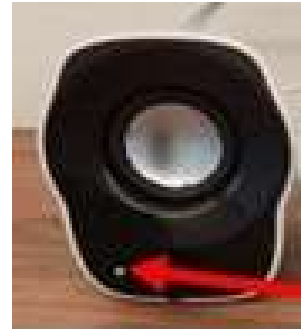
the speaker's USB  
power LED



# Related Work

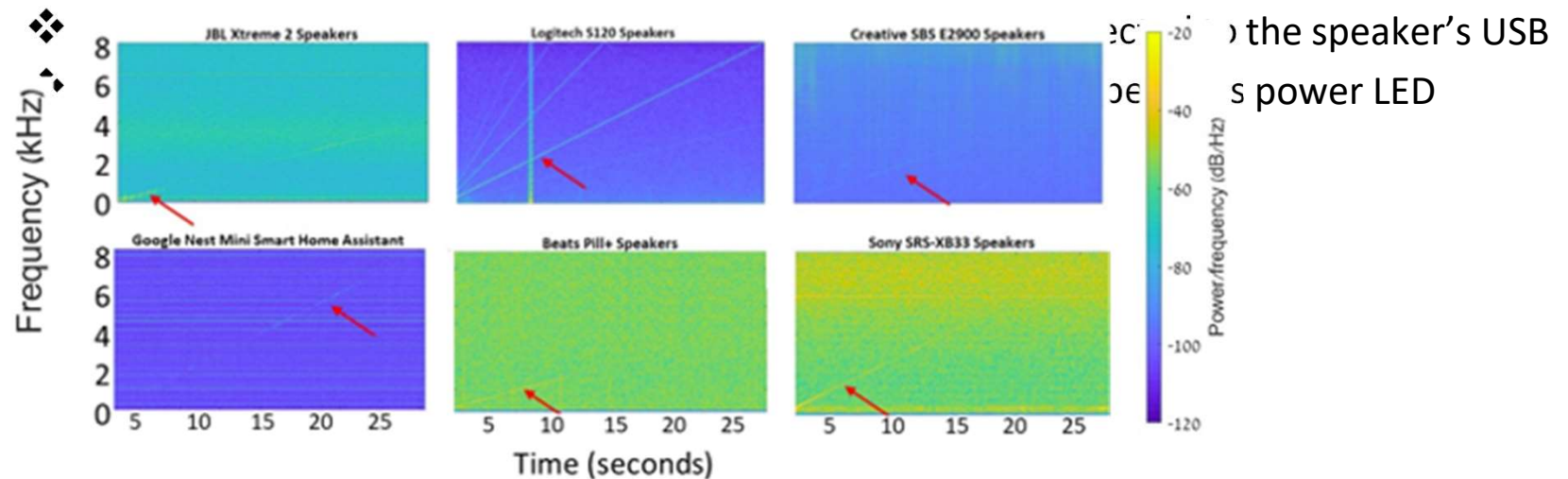
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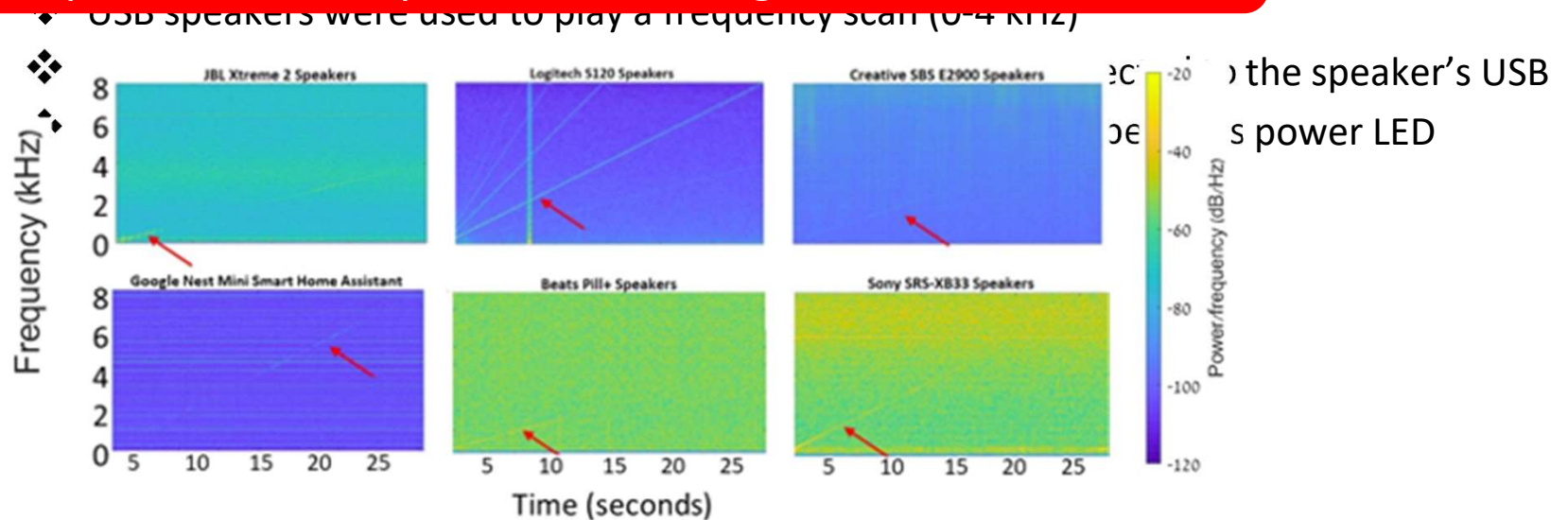
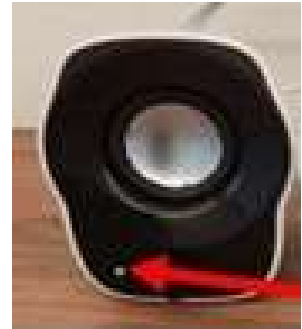


# Related Work

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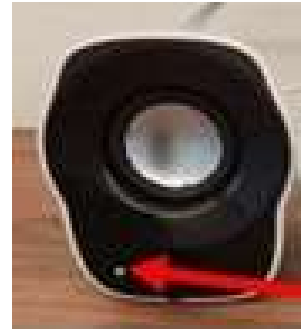
Why can the intensity of a speaker's power LED be used to recover speech?

The intensity of the power LED of various speakers correlates with the power consumption in the range of 0-4 kHz.



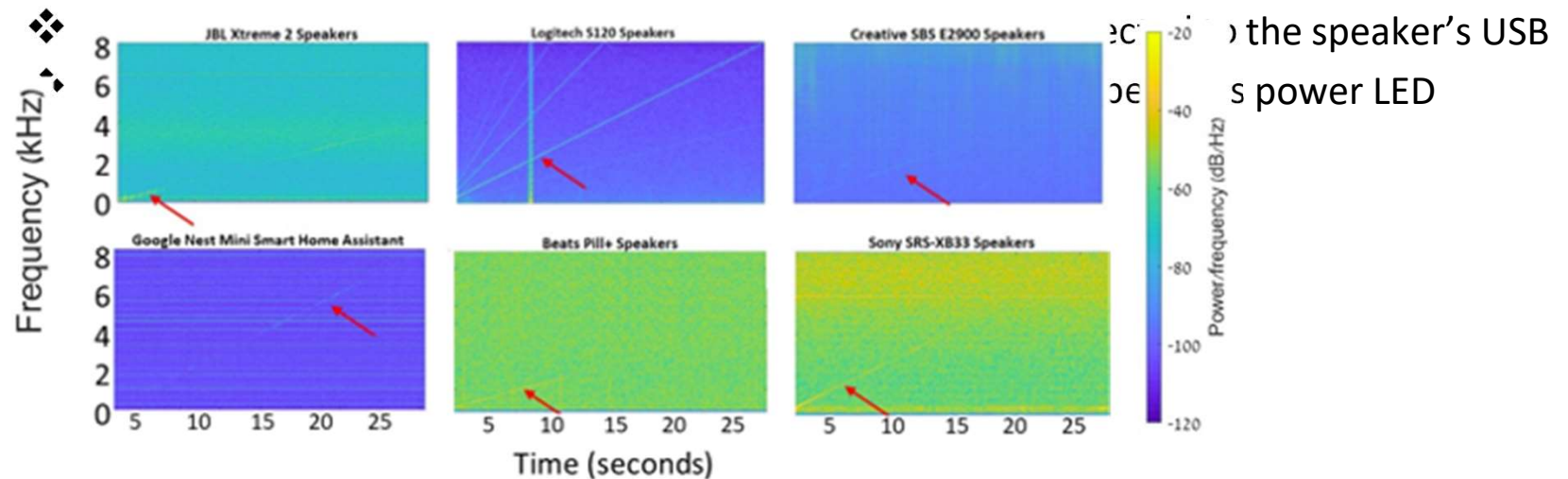
# Related Work

Why? In some electrical circuits, the integrated power LED is connected directly to the power line, and dedicated means of decoupling the optical and power correlation are either not integrated or are integrated into the circuits but ineffective.



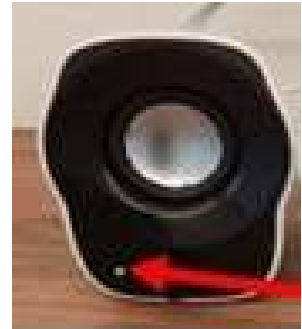
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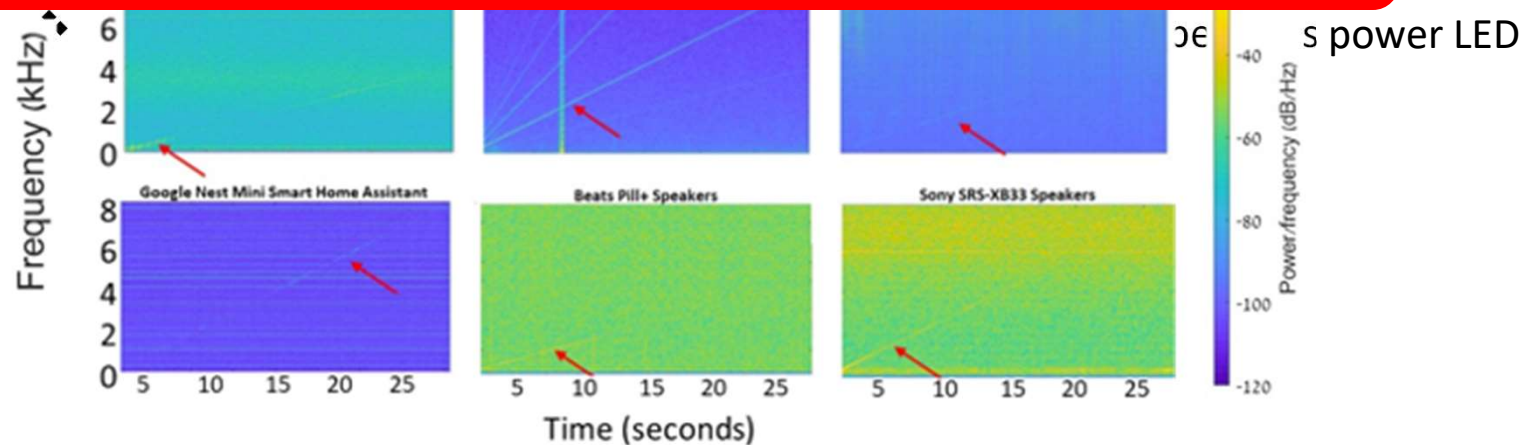
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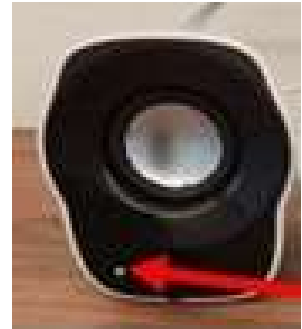
As a result, the power consumption of the device (power supply to the integrated power LED) affects the intensity of the LED.





# Related Work

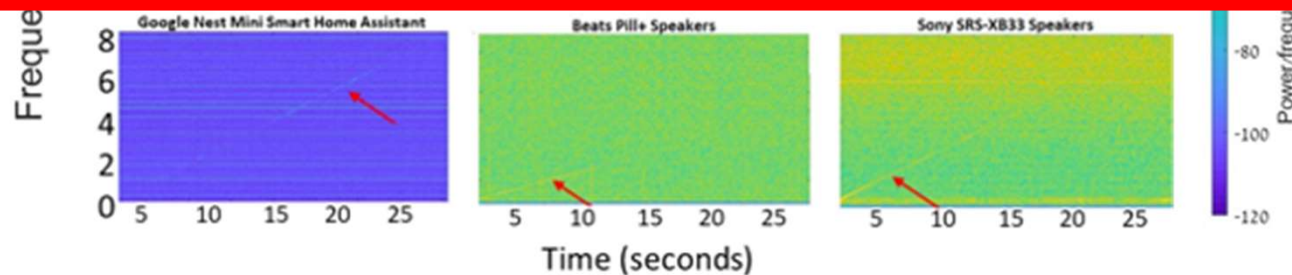
Why? In some electrical circuits, the integrated power LED is connected directly to the power line, and dedicated means of decoupling the optical and power correlation are either not integrated or are integrated into the circuits but ineffective.



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As a result, the power consumption of the device (power supply to the integrated power LED) affects the intensity of the LED.

The optical measurements reflect the power consumption of the device.

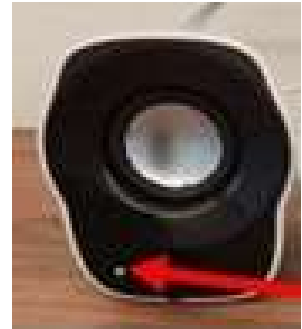


speaker's USB  
over LED



# Related Work

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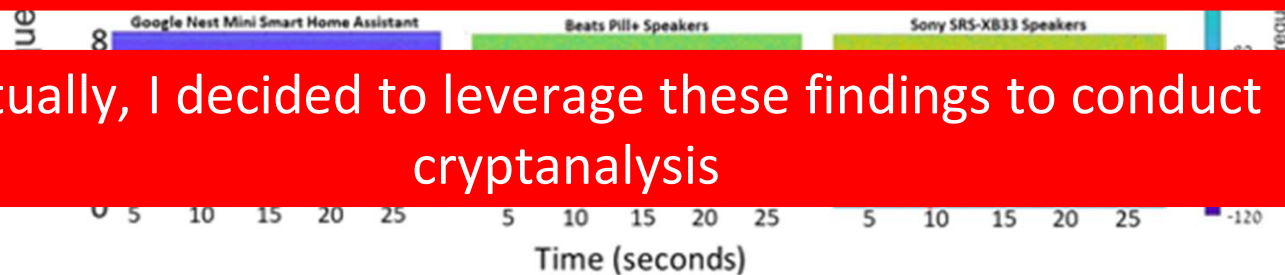


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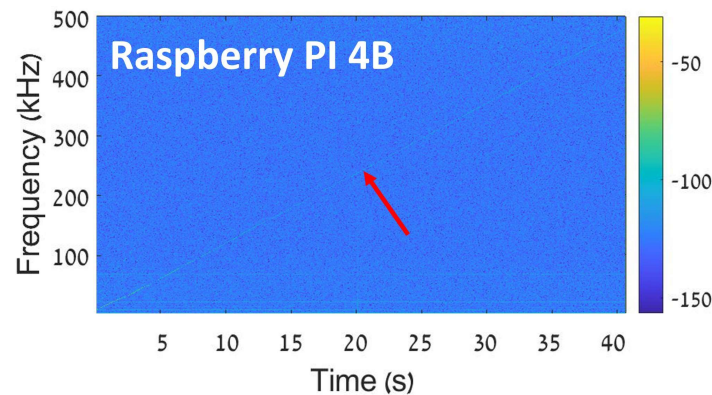
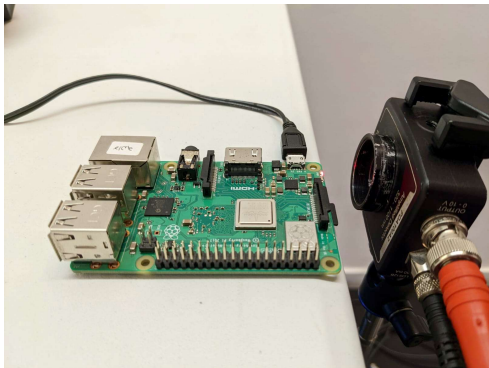
Eventually, I decided to leverage these findings to conduct cryptanalysis



speaker's USB  
over LED

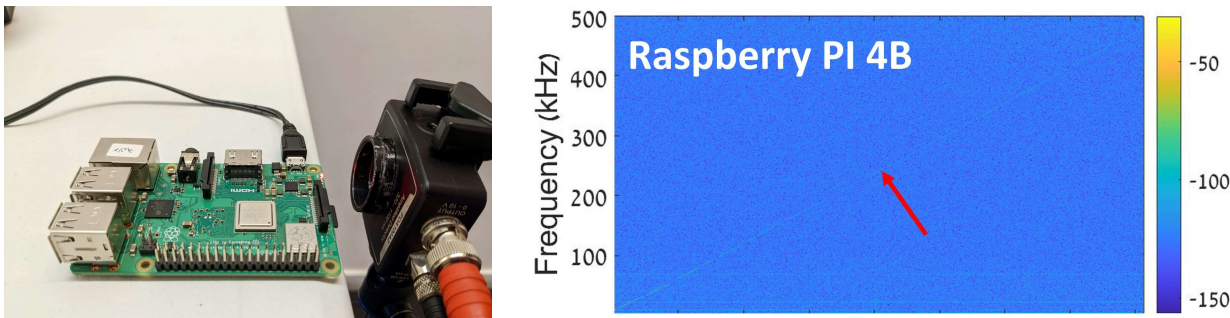
# Related Work

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  - ❖ Key recovery from a device's power LED using a photodiode



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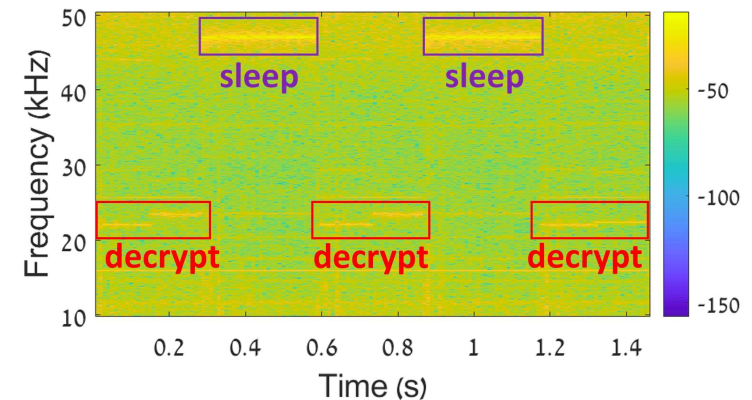
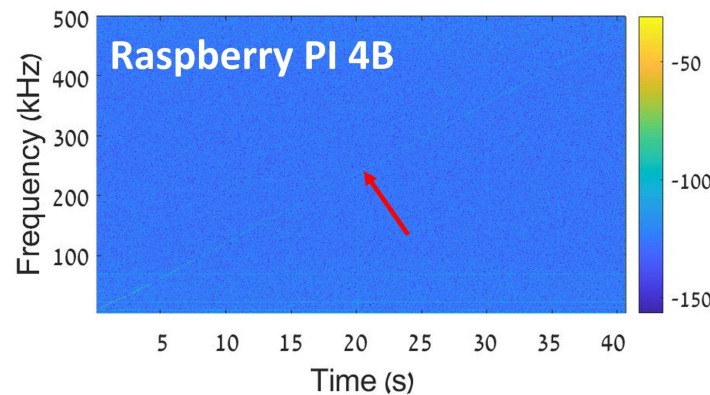
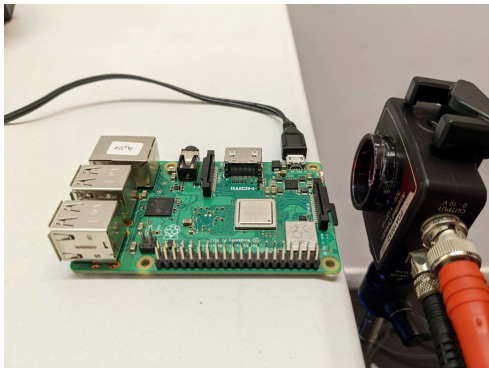
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The intensity of the power LED of various devices correlates with the power consumption of the device in a wide spectrum (0-500 kHz).

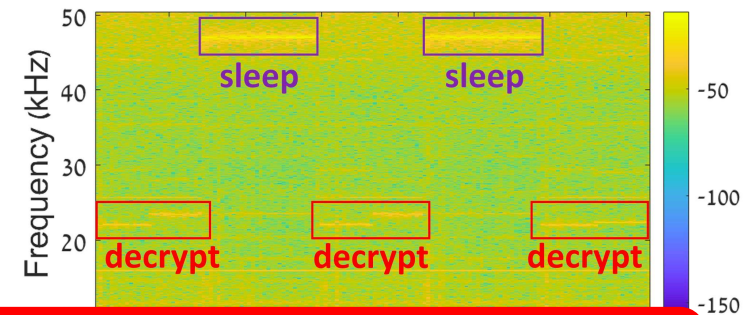
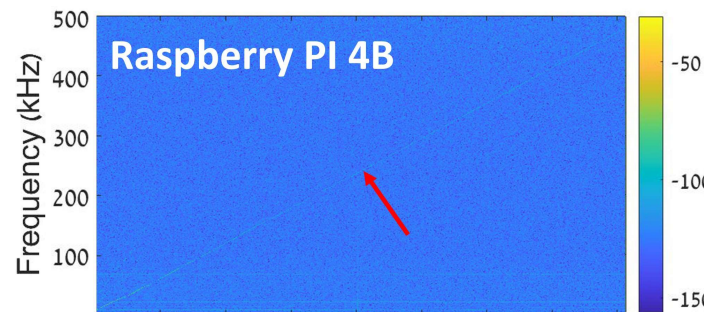
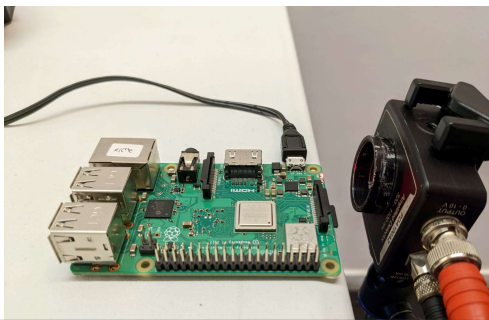
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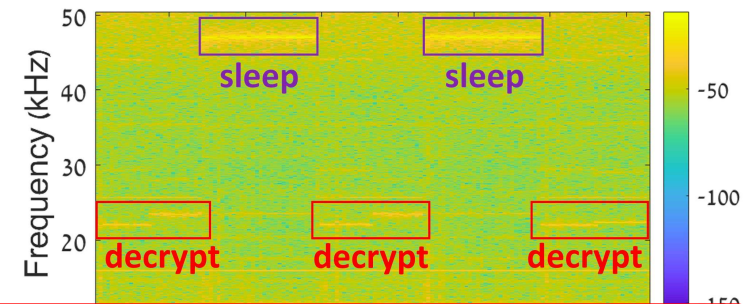
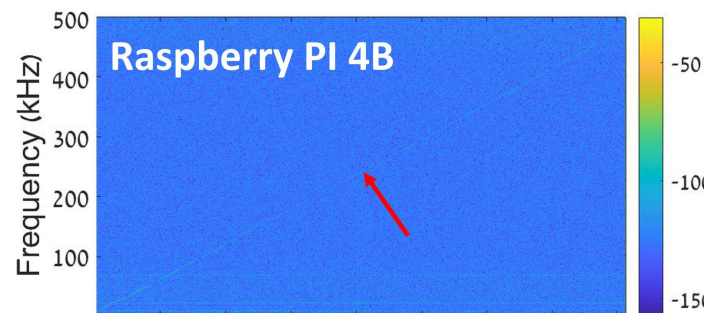
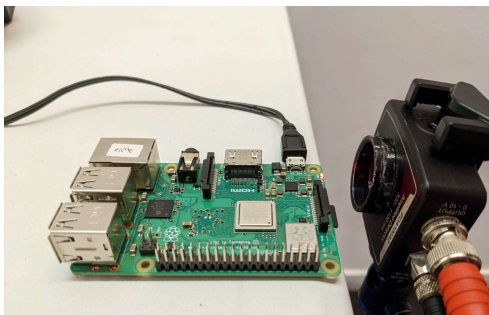


By obtaining optical measurements from a device's power LED, we could detect the beginning and end of the cryptographic operations performed by the CPU.



# Related Work

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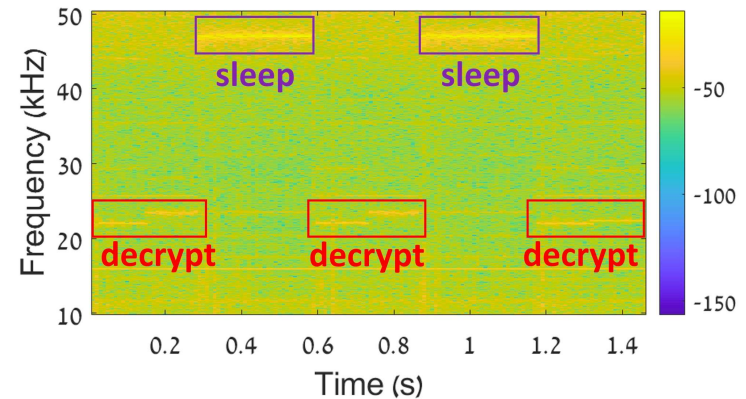
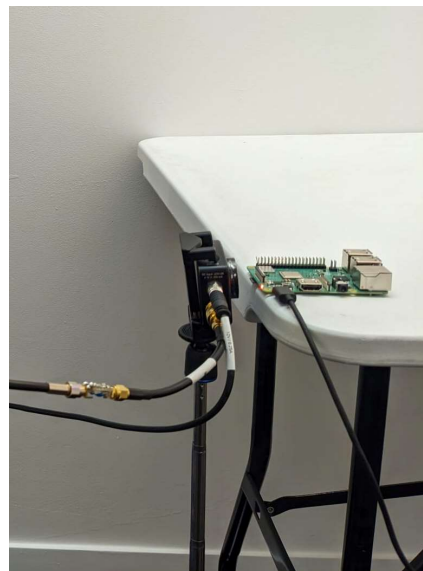
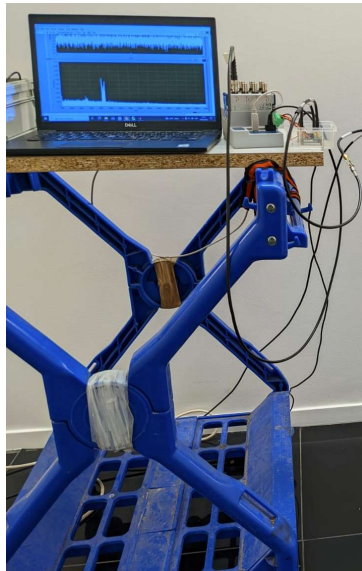


We applied cryptanalytic timing attacks using optical measurements obtained from a device's power LED by a photodiode and recovered RSA, ECDSA, and SIKE keys.

# Related Work

## ❖ 2023 - Optical Cryptanalysis (Nassi et al.) at CCS 2023

### ❖ Key recovery from a device's power LED using a photodiode

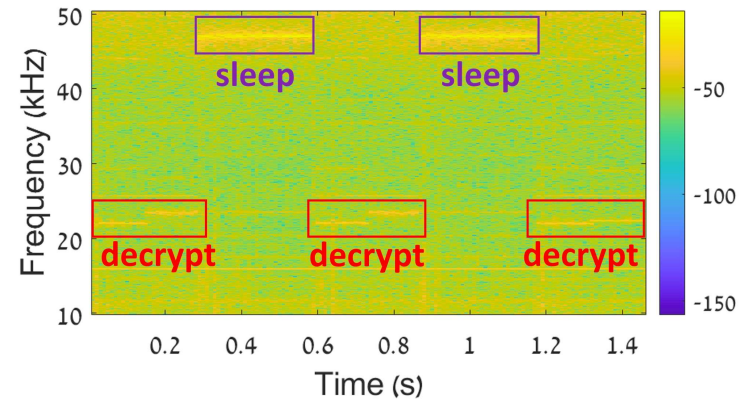
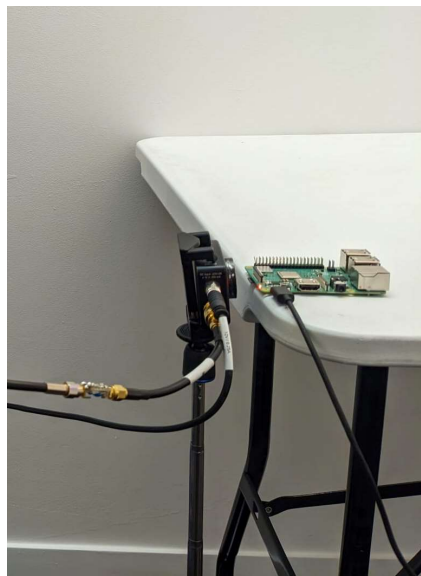
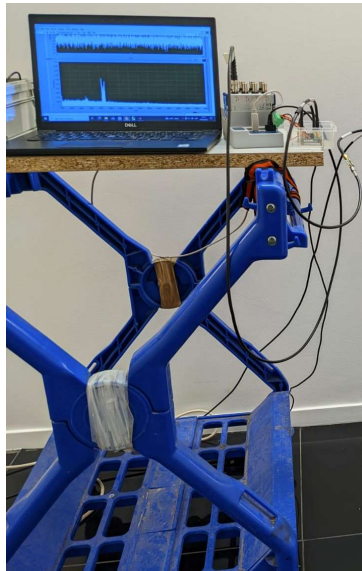


Primary disadvantage: Photodiodes are not commonly used sensors, and attackers must connect them to a dedicated ADC in order to obtain optical measurements.

# Related Work

❖ 2023 - Optical Cryptanalysis (Nassi et al.) at CCS 2023

❖ Key recovery from a device's power LED using a photodiode



Solution: Let's perform cryptanalysis by using ubiquitous video cameras to obtain optical measurements instead of photodiodes.



# Threat Models

Objective: Performing cryptanalysis (recovering secret keys) from a device's power LED using video cameras instead of photodiodes.

Close Video Acquisition

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## Close Video Acquisition

- Uses a smartphone's video camera



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- Targets any type of power LED



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## Over the Internet Video Acquisition

- Uses an Internet-connected video camera



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- Uses an Internet-connected video camera
- Targets Type 2 Power LED (commonly used in smartcard readers)





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## Over the Internet Video Acquisition

- Uses an Internet-connected video camera
- Targets Type 2 Power LED (commonly used in smartcard readers)
- Can be applied remotely over the Internet (using a hijacked video camera)



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## Over the Internet Video Acquisition

- Uses an Internet-connected video camera
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So far, we've used photodiodes, which are analog sensors that can be sampled at a few GHz.

(using a hijacked video camera)



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Video cameras can record 60 or 120 frames per second.



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A sampling rate of 60 frames per second is not sufficient to perform cryptanalysis.



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Video cameras can record 60 or 120 frames per second.

A sampling rate of 60 frames per second is not sufficient to perform cryptanalysis.



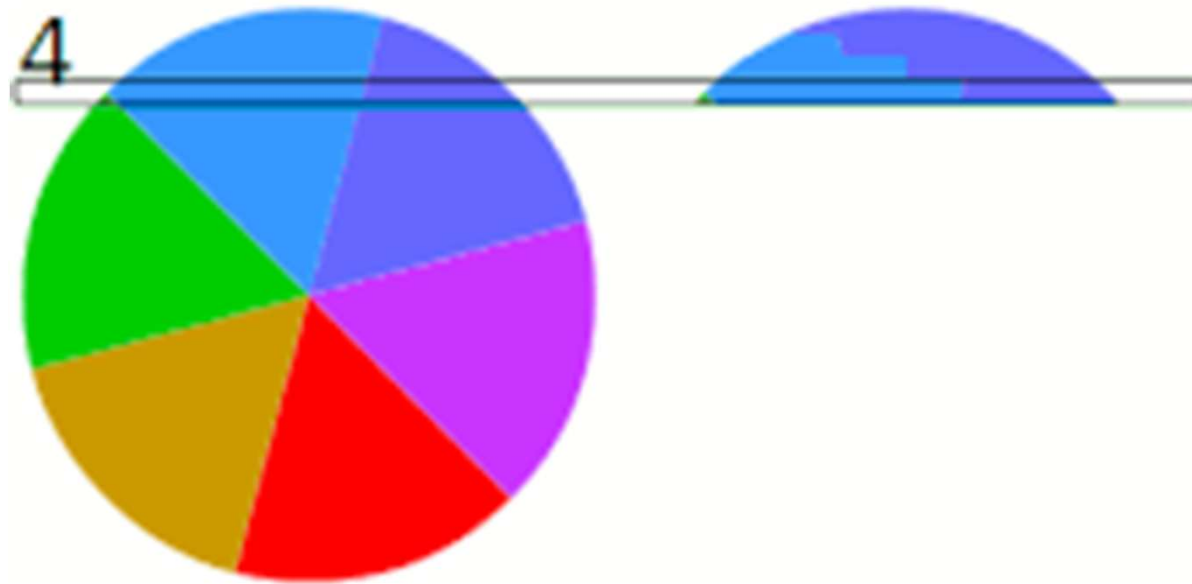
Or is it?



# Rolling Shutter

1. We used to think that a picture is the result of one atomic snapshot taken by the video camera.
2. 99.9% of video cameras use a rolling shutter to record a video.
  - A single picture is the result of multiple snapshots taken at different times by the video camera.
  - A video camera uses a rolling shutter to scan an object vertically or horizontally and combines the scanned pieces into a single frame.
  - Only a few rows are captured atomically each time, and the entire picture consists of multiple snapshots of an object.

# Rolling Shutter



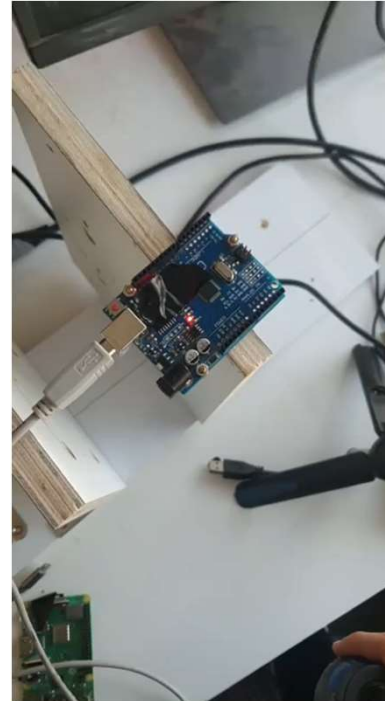
Distortions appear in pictures of fast-moving objects. Such distortions should not appear if the picture was taken using one atomic snapshot of the object.



# Rolling Shutter

## Experiment

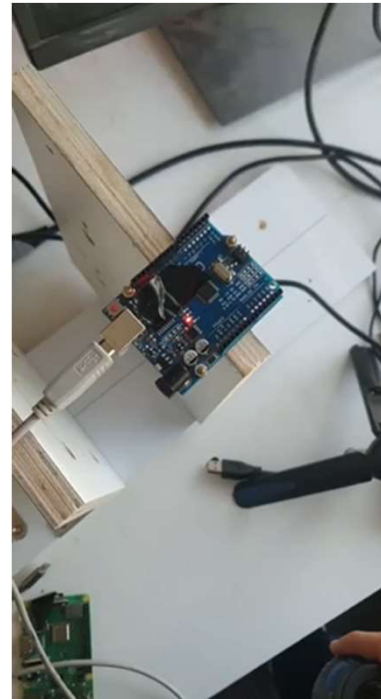
1. We programmed an Arduino to turn its power LED on and off every 250 microseconds.



# Rolling Shutter

## Experiment

1. We programmed an Arduino to turn its power LED on and off every 250 microseconds.

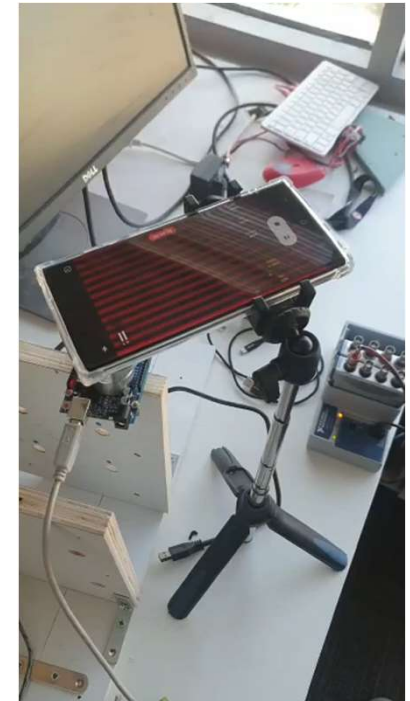
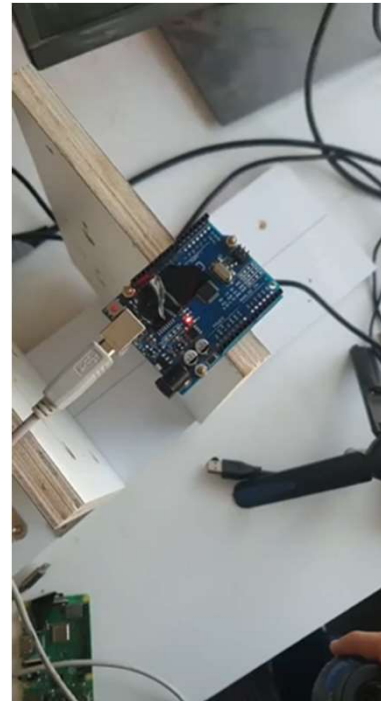


The flickering LED cannot be seen at 4 kHz, because the sampling rate is 60 FPS.

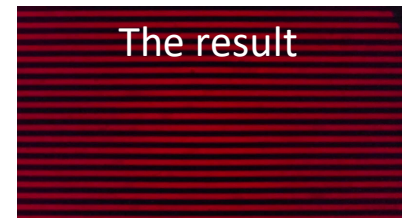
# Rolling Shutter

## Experiment

1. We programmed an Arduino to turn its power LED on and off every 250 microseconds.
2. An extra lens was used to direct the camera of a Samsung Galaxy S22 Ultra to the Arduino's power LED, so that the view of the LED fills the entire frame.



The result

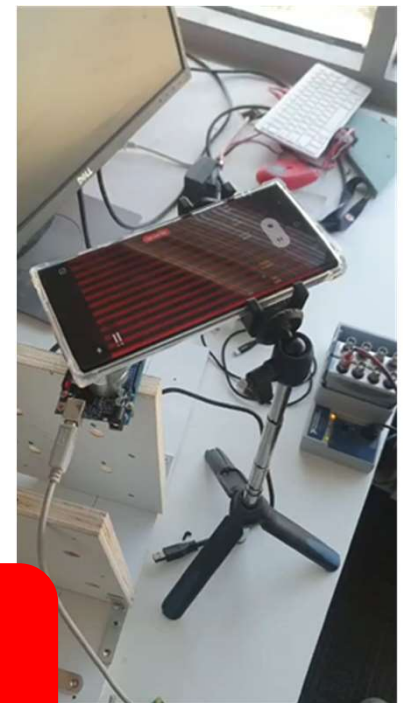
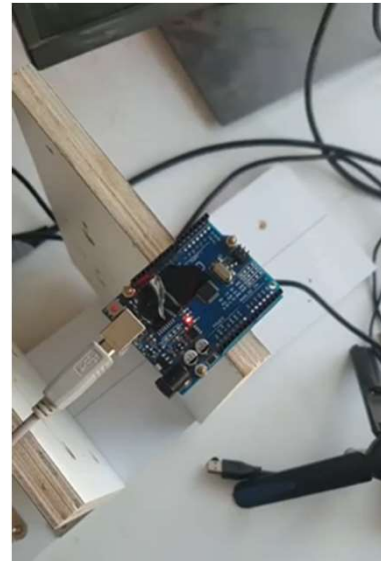


# Rolling Shutter

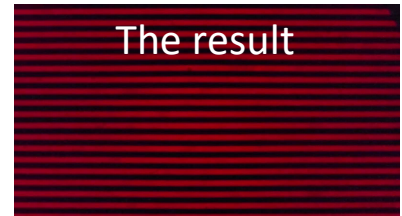
## Experiment

1. We programmed an Arduino to turn its power LED on and off every 250 microseconds.
2. An extra lens was used to direct the camera of a Samsung Galaxy S22 Ultra to the Arduino's power LED, so

By filling the frame with the view of the LED, we exploit a video camera's rolling shutter to increase the number of measurements obtained at different times.



The result



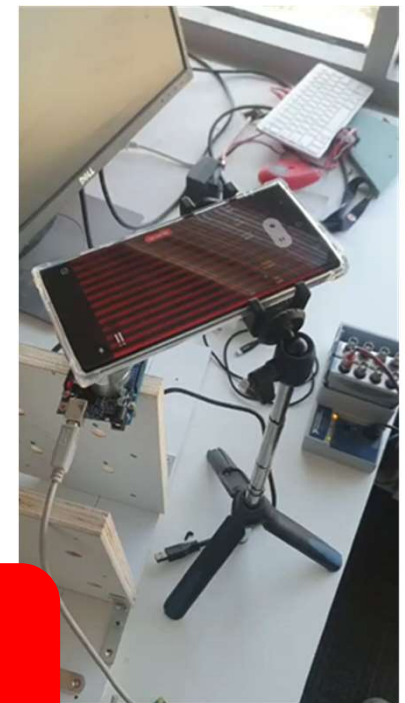
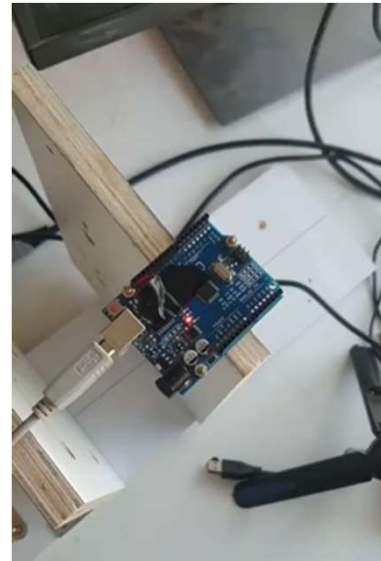
# Rolling Shutter

## Experiment

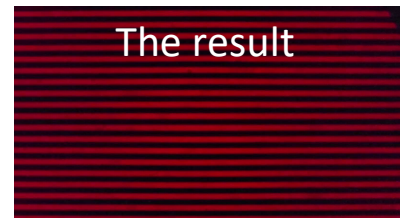
1. We programmed an Arduino to turn its power LED on and off every 250 microseconds.
2. An extra lens was used to direct the camera of a Samsung Galaxy S22 Ultra to the Arduino's power LED, so

By filling the frame with the view of the LED, we exploit a video camera's rolling shutter to increase the number of measurements obtained at different times.

This allows us to detect the 4 kHz flicker



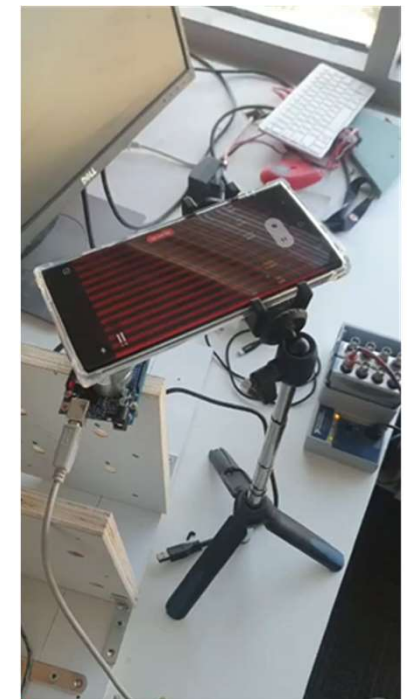
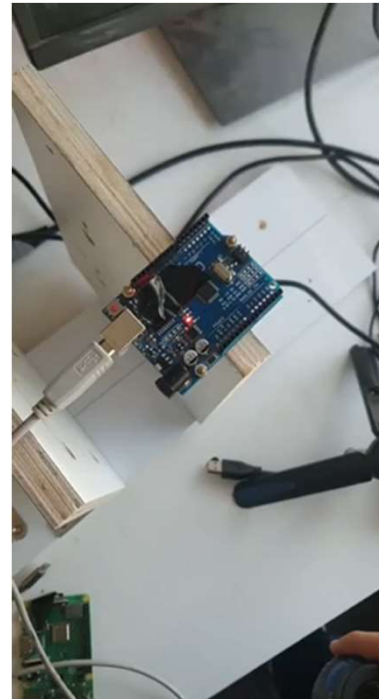
The result



# Rolling Shutter

## Experiment

1. We programmed an Arduino to turn its power LED on and off every 250 microseconds.
2. An extra lens was used to direct the camera of a Samsung Galaxy S22 Ultra to the Arduino's power LED, so that the view of the LED fills the



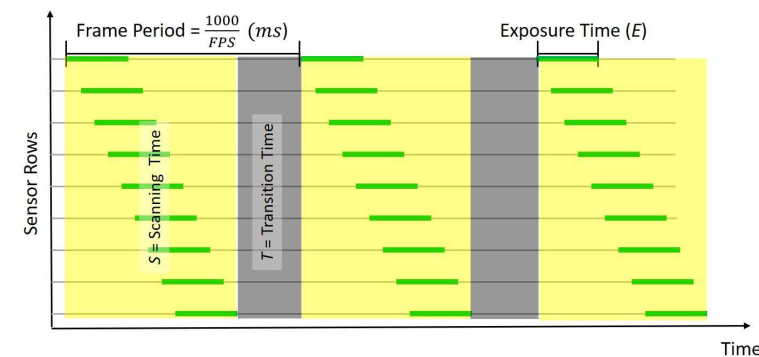
Theoretically, by exploiting a video camera's rolling shutter, we can increase the sampling rate by three orders of magnitude: from the FPS rate (60 measurements per second) to the rolling shutter rate (60K).

The result



# Rolling Shutter

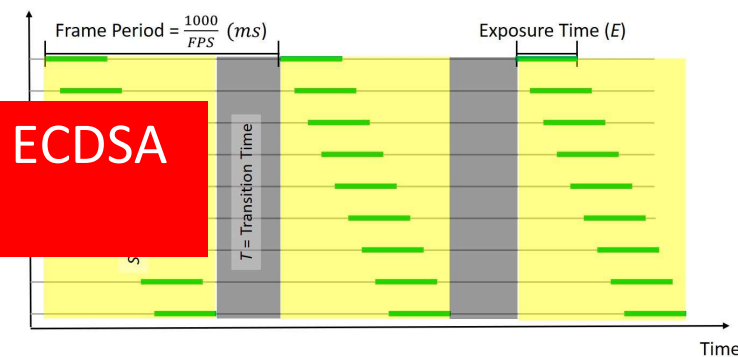
1. In reality, there is a delay (transition time) between consecutive frames.
2. The transition between frames is a period of time during which no object is captured by the video camera.
3. This should be taken into account when analyzing the video footage of the power LED.



# Rolling Shutter

1. In reality, there is a delay (transition time) between consecutive frames.
2. The transition between frames is a period of time during which no object is captured by the video camera.
3. This should be taken into account when analyzing the video footage of the power LED.

Now, I will let Ben to discuss the ECDSA key recovery.





# Recovering ECDSA Keys

## Minerva Attack

1. The attack was published in 2020.
2. The authors found that the implementation of ECDSA on SECP256R1 (with random nonces and hash function SHA-256) in many cryptographic libraries applies some runtime optimizations.
  - The number of iterations of the main loop is determined by the number of leading zeroes in the nonce.
  - It creates a time dependency between the number of leading zeros in the nonce and the execution time of the main loop.

---

### Algorithm 1 Libcrypt's Double-and-Always-Add Scalar Multiplication

---

**Input:** A positive scalar  $k = k_{n-1} \cdots k_0$  and an elliptic-curve point  $G$ .  
**Output:**  $[k]G$ .

```
1: procedure DOUBLE-ALWAYS-ADD( $k, G$ )  
2:    $Q \leftarrow O$  ▷  $O$  is identity element  
3:    $t \leftarrow \text{truncate-leading-zeros}(k)$   
4:   for  $i \leftarrow t - 1$  to 0 do  
5:      $Q \leftarrow 2Q$  ▷ always double  $Q$   
6:      $T \leftarrow Q + G$   
7:      $Q, T \leftarrow \text{ct-swap}(Q, T, k_i)$  ▷ swap if  $k_i = 1$   
8:   return  $Q$ 
```

---

# Recovering ECDSA Keys

## Minerva Attack

1. The attack was published in 2020.

2. Attackers can determine the number of leading zeros in the nonce by analyzing the ECDSA signing time.

SHA-256) in many cryptographic libraries applies

Measuring the signing time -> estimating the number of iterations -> determining the number of leading zeroes in the nonce.

is determined by the number of the leading

The Minerva paper (2020) introduced a new timing attack to recover the complete ECDSA key based on the number of leading zeros in the nonces of a set of signatures (~4,000 signatures are needed).

execution time of the main loop.

---

**Algorithm 1** Libgcrypt's Double-and-Always-Add Scalar Multiplication

---

**Input:** A positive scalar  $k = k_{n-1} \cdots k_0$  and an elliptic-curve point  $G$ .

**Output:**  $[k]G$ .

*Libgcrypt procedure DOUBLE\_ALWAYS\_ADD( $k, G$ )*

*1.  $Q \leftarrow G$  //  $G$  is identity element*

*2.  $Q \leftarrow 2Q$  // always double  $Q$*

*3.  $Q \leftarrow Q + G$  //  $\triangleright$  swap if  $k_i = 1$*

*4.  $Q \leftarrow 2Q$*

*5.  $Q \leftarrow Q + G$*

*6.  $Q \leftarrow 2Q$*

*7.  $Q \leftarrow Q + G$*

*8.  $Q \leftarrow 2Q$*

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*215.  $Q \leftarrow Q + G$*

# Recovering ECDSA Keys

## Minerva Attack

1. The attack was published in 2020.
2. The authors found that the implementation of SECP256R1 (with random ECDSA nonces and

**Algorithm 1** Libcrypt's Double-and-Always-Add Scalar Multiplication

**Input:** A positive scalar  $k = k_{n-1} \dots k_0$  and an elliptic-curve point  $G$ .  
**Output:**  $[k]G$ .

```
1: procedure DOUBLE-ALWAYS-ADD( $k, G$ )  
2:    $Q \leftarrow O$  ▷  $O$  is identity element  
3:    $t \leftarrow \text{truncate-leading-zeros}(k)$   
4:   for  $i \leftarrow t - 1$  to 0 do  
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6:      $T \leftarrow Q + G$   
7:      $Q, T \leftarrow \text{ct-swap}(Q, T, k_i)$  ▷ swap if  $k_i = 1$   
8:   return  $Q$ 
```

The original Minerva attack required the attackers to obtain timing measurements from the CPU of the host machine (i.e., the attackers needed to compromise the host machine).

- The number of iterations of the main loop

The authors were unable to demonstrate the attack remotely over the Internet due to the noise added by the network latency (because the Minerva attack is very sensitive to errors in the signing measurements).

- This creates a time dependency between the number of leading zeros in the nonce and the execution time of the main loop.

# Recovering ECDSA Keys

## Algorithm 1 Libcrypt's Double-and-Always-Add Scalar Multiplication

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```

Signatures + Execution/CPU time of a set of ECDSA signatures

0xD10B.....0F, 143.2 ms

.....  
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Minerva Attack

Black Box

ECDSA Key



Minerva recovers ECDSA key from a set of ECDSA signatures with associated signing times

# Recovering ECDSA Keys

---

**Algorithm 1** Libcrypt's Double-and-Always-Add Scalar Multiplication

---

**Input:** A positive scalar  $k = k_{n-1} \cdots k_0$  and an elliptic-curve point  $G$ .

**Output:**  $[k]G$ .

1: **procedure** DOUBLE-ALWAYS-ADD( $k, G$ )

2:    $Q \leftarrow O$  ▷  $O$  is identity element

3:    $t \leftarrow \text{truncate-leading-zeros}(k)$

4:   **for**  $i \leftarrow t - 1$  **to** 0 **do**

5:      $Q \leftarrow 2Q$  ▷ always double  $Q$

6:      $T \leftarrow Q + G$

7:      $Q, T \leftarrow \text{ct-swap}(Q, T, k_i)$  ▷ swap if  $k_i = 1$

8:   **return**  $Q$

---

Signatures + Execution/CPU time of a  
set of ECDSA signatures

0xD10B.....0F, 143.2 ms

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Minerva Attack

Black Box

ECDSA Key



The details related to the implementation of the blackbox appear in the paper

# Recovering ECDSA Keys

---

**Algorithm 1** Libcrypt’s Double-and-Always-Add Scalar Multiplication

**Input:** A positive scalar  $k = k_{n-1} \cdots k_0$  and an elliptic-curve point  $G$ .

**Output:**  $[k]G$ .

1: **procedure** DOUBLE-ALWAYS-ADD( $k, G$ )2:  $Q \leftarrow 0$ 

▷  $O$  is identity element

3:  $t \leftarrow \text{truncate-leading-zeros}(k)$ 

```

4:   for  $i \leftarrow t - 1$  to 0 do

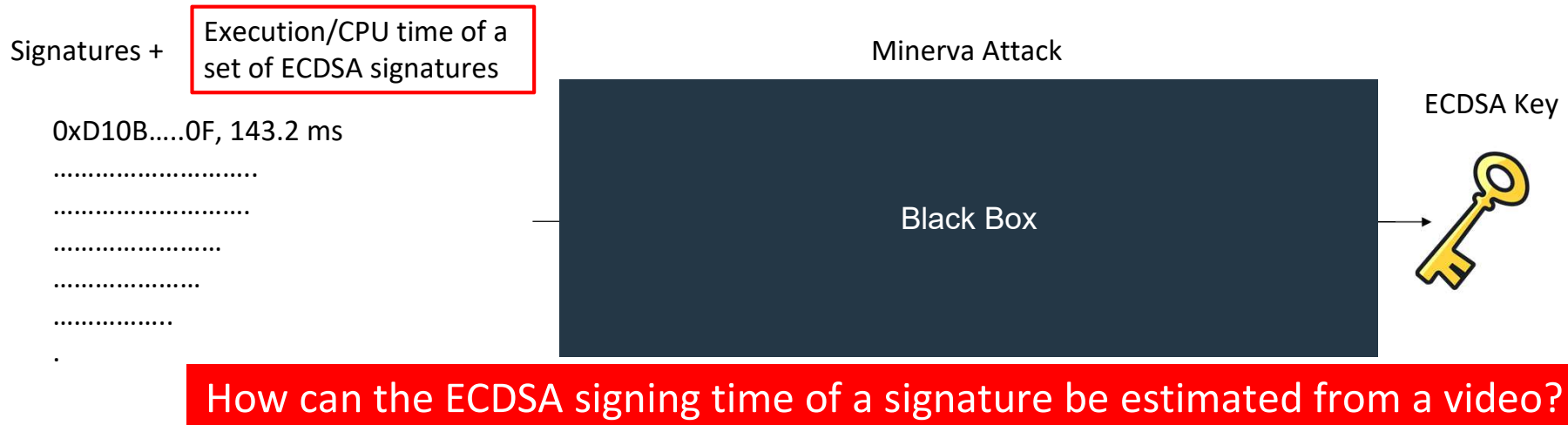
```

5:  $Q \leftarrow 2Q$ 

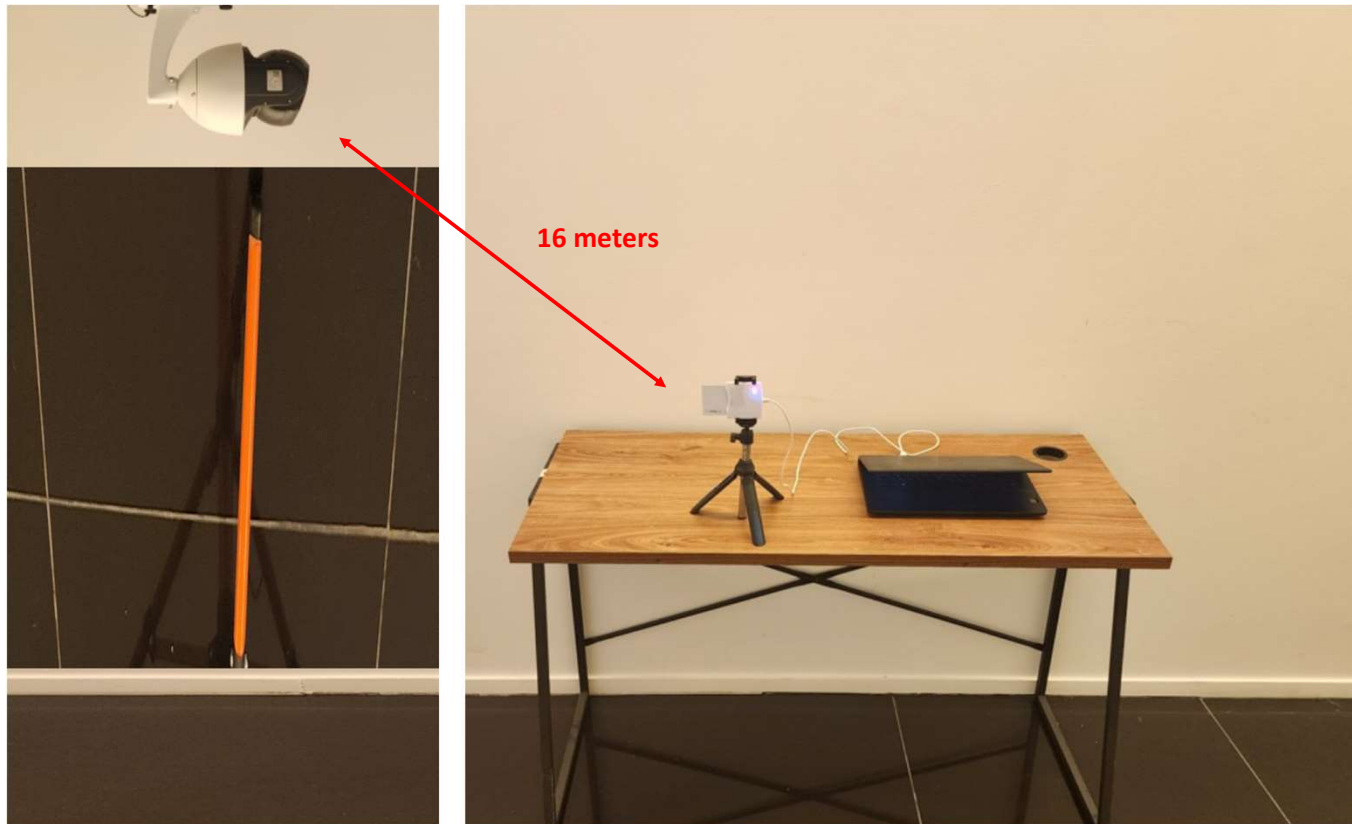
▷ always double  $Q$

6:  $T \leftarrow Q + G$ 7:  $Q, T \leftarrow \text{ct-swap}(Q, T, k_i)$ 

▷ swap if  $k_i = 1$

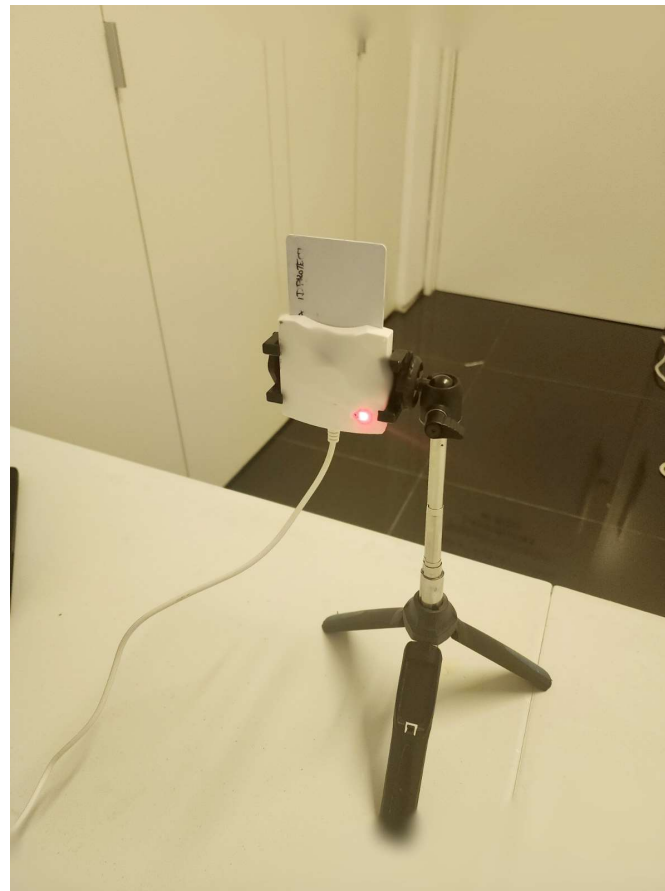
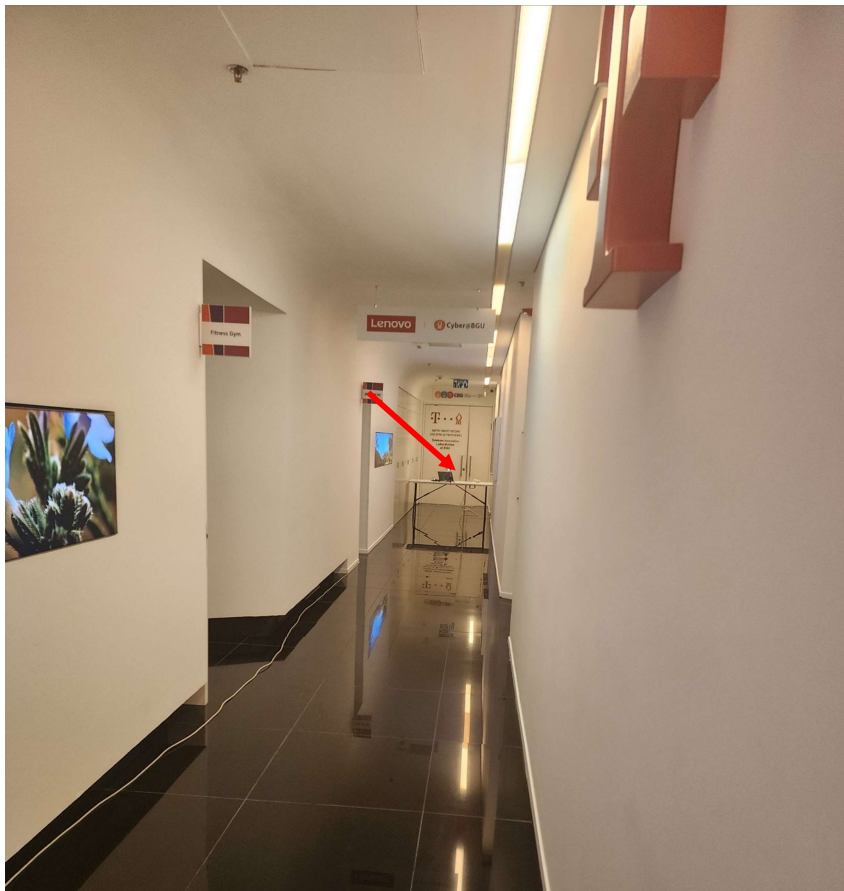
8: **return**  $Q$ 

# Recovering ECDSA Keys



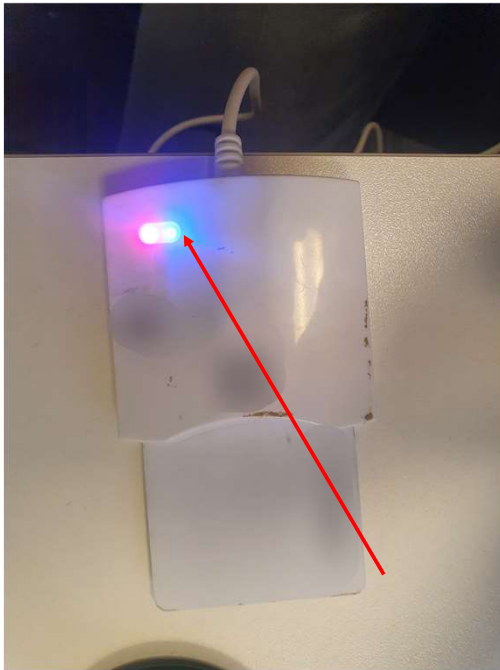
Objective: Recovering a 256-bit ECDSA key from a smartcard using video footage of the power LED of the smartcard reader obtained by an Internet-connected video-camera.

# Recovering ECDSA Keys





# Recovering ECDSA Keys

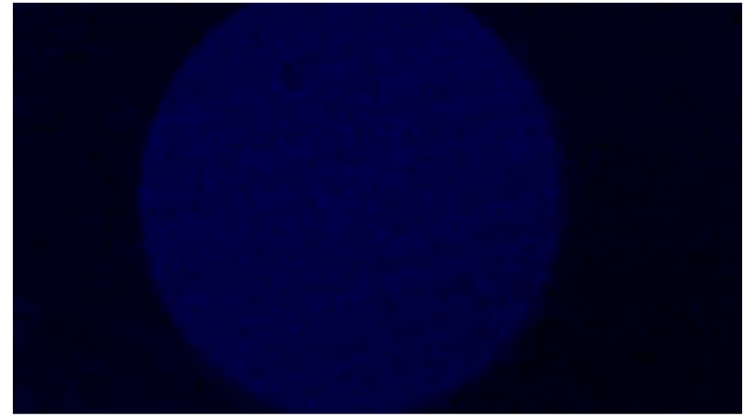
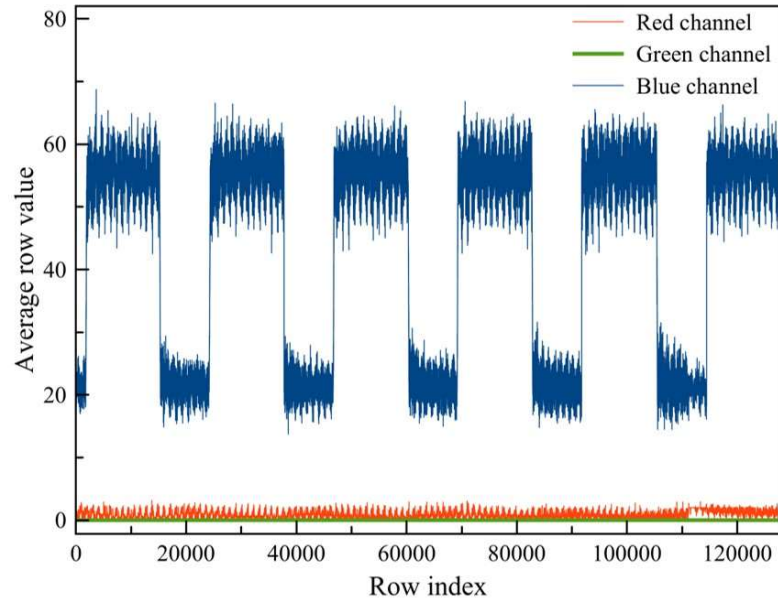


Right LED  
Blue = idle  
Off = sign

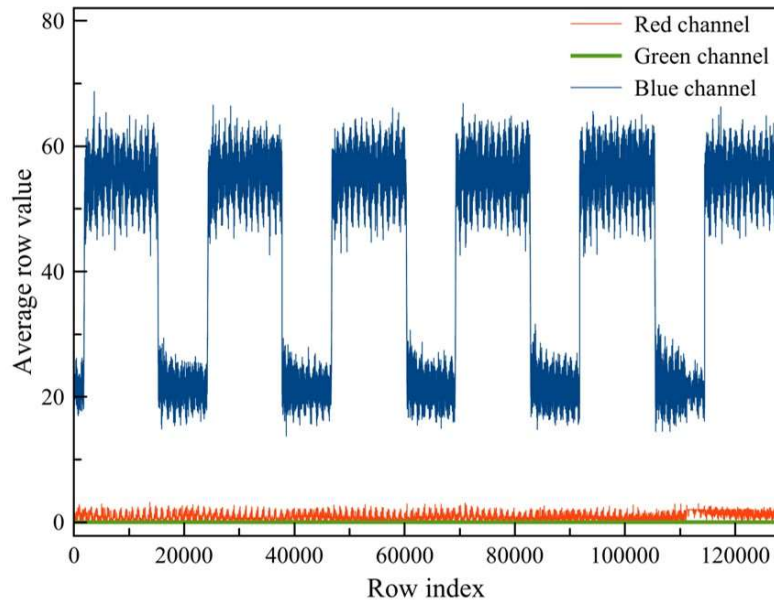


Speed x4

# Recovering ECDSA Keys



# Recovering ECDSA Keys



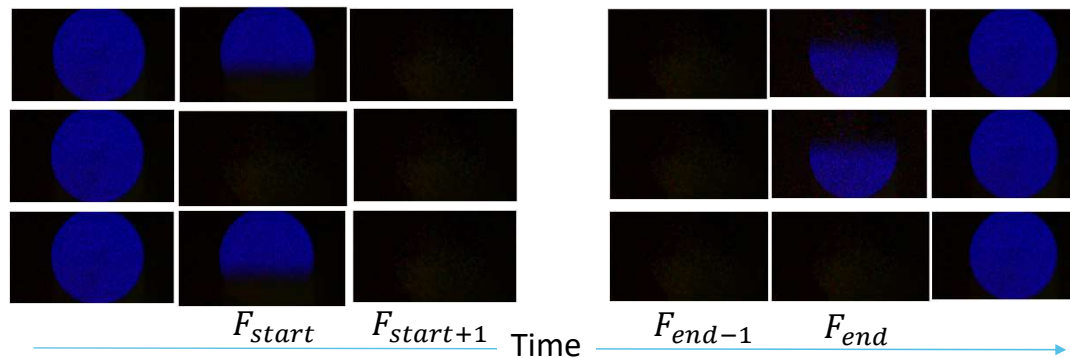
By averaging each row in a frame to a single value and arranging them in a time series, we can distinguish between sign operations (the power LED is off) and idle operations (the color of the power LED is blue) based on the blue channel.

But how can we calculate the signing time?

# Recovering ECDSA Keys

Calculating the ECDSA signing time

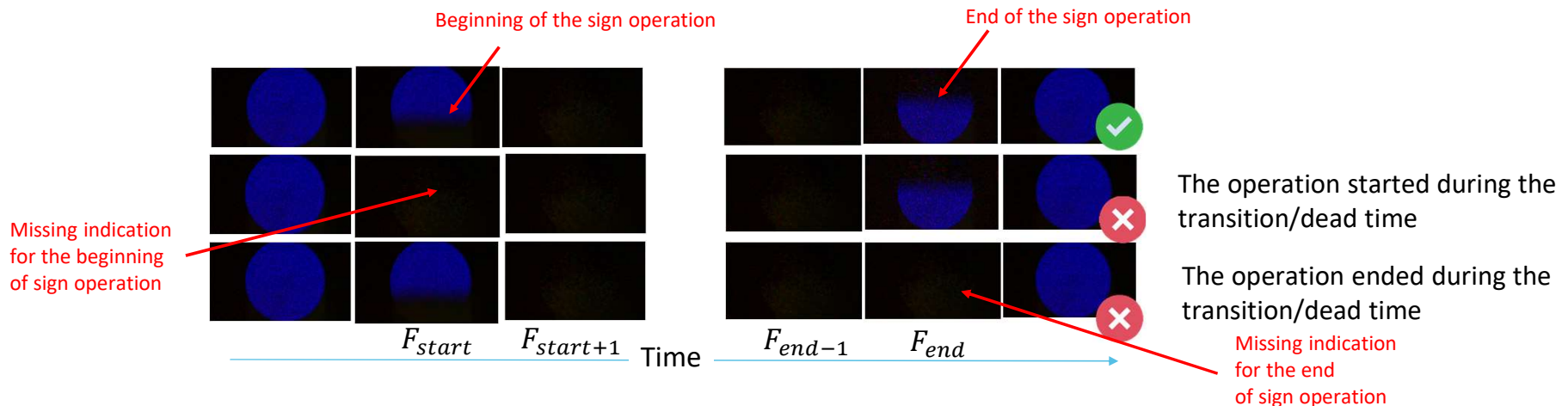
1. We extracted the series of frames associated with each signature from the video.



# Recovering ECDSA Keys

Calculating the ECDSA signing time

1. We extracted the series of frames associated with each signature from the video.
2. We kept series that contained the indication of the switch between the beginning and the end of signing, i.e. that the beginning and the end of the signing were captured in the frames (we filtered series in which one of the indications is missing).



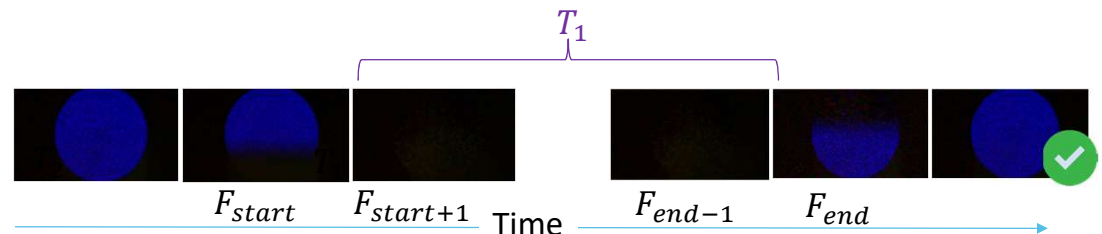
# Recovering ECDSA Keys

Calculating the ECDSA signing time

1. We extracted the series of frames associated with each signature from the video.
2. We kept series that contained the indication of the switch between the beginning and the end of signing
3. We calculated  $T_1$  = the execution time of all of the full black frames ( $F_{start+1} \dots F_{end+1}$ ).

$$number\_of\_full\_frames = 1 + F_{end-1} - F_{start}$$

$$T_1 = \frac{1000}{FPS} * number\_of\_full\_frames$$

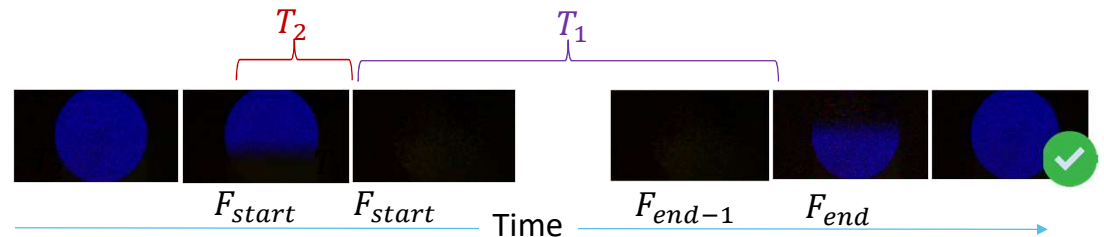


# Recovering ECDSA Keys

Calculating the ECDSA signing time

1. We extracted the series of frames associated with each signature from the video.
2. We kept series that contained the indication of the switch between the beginning and the end of signing
3. We calculated  $T_1$  = the execution time of all of the full red/black frames ( $F_{start+1} \dots F_{end+1}$ ).
4. We calculated  $T_2$  = the execution time of the first frame in which the color changed ( $F_{start}$ ).

$$T_2 = \frac{\text{Number\_of\_Relevant\_Rows\_in\_}F_{start}}{\text{Total\_Rows\_in\_Frame}} * S + T$$

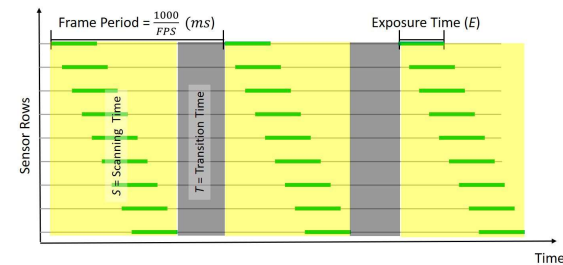


# Recovering ECDSA Keys

Calculating **S** and **T**:

Given a video camera with a known **FPS** and a frame period  $P_{(\text{millisec})} = 1000/\text{FPS} = \mathbf{S} + \mathbf{T}$  that consists of a scanning time **S** and a transition time **T** between frames

1. Measure **S** using a flicker
2. Calculate  $\mathbf{T} = (1000/\text{FPS}) - \mathbf{S}$



- 39 changes in the color of the flicker
- that was set to work at a frequency of 4000 Hz
- $39 \times 250 \text{ microseconds} = 9.75$

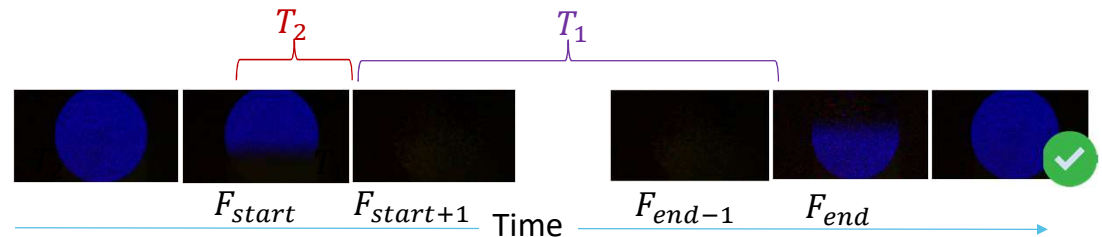


# Recovering ECDSA Keys

Calculating the ECDSA signing time

1. We extracted the series of frames associated with each signature from the video.
2. We kept series that contained the indication of the switch between the beginning and the end of signing
3. We calculated  $T_1$  = the execution time of all of the frames ( $F_{start+1} \dots F_{end+1}$ ).
4. We calculated  $T_2$  = the execution time of the first frame in which the color changed ( $F_{start}$ ).

$$T_2 = \frac{\text{Number\_of\_Relevant\_Rows\_in\_}F_{start}}{\text{Total\_Rows\_in\_Frame}} * S + T$$

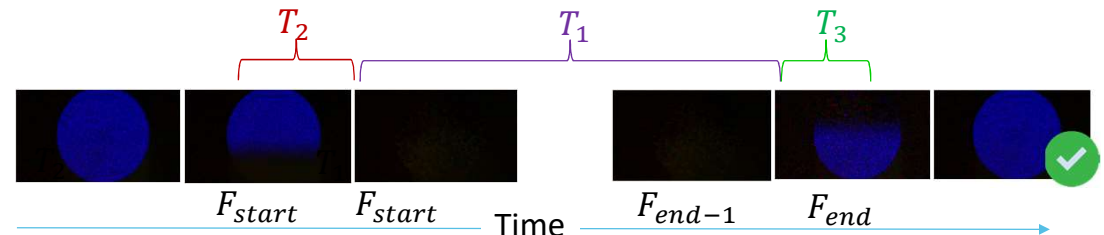


# Recovering ECDSA Keys

Calculating the ECDSA signing time

1. We extracted the series of frames associated with each signature from the video.
2. We kept series that contained the indication of the switch between the beginning and the end of signing
3. We calculated  $T_1$  = the execution time of all of the frames ( $F_{start} \dots F_{end+1}$ ).
4. We calculated  $T_2$  = the execution time of the first frame in which the color changed ( $F_{start}$ ).
5. We calculated  $T_3$  = the execution time of the last frame in which the color changed ( $F_{end}$ ).

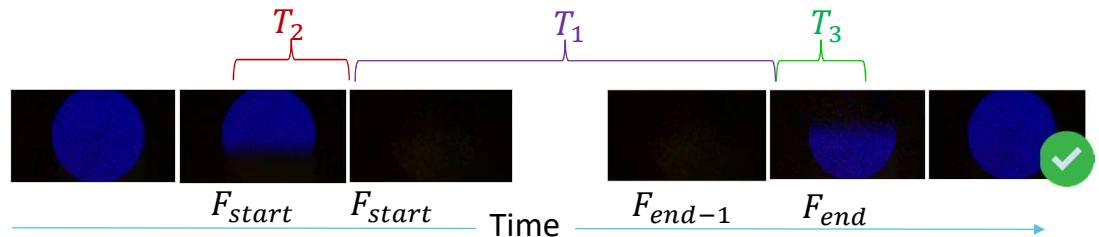
$$T_3 = \frac{\text{Number\_of\_Relevant\_Rows\_in\_}F_{end}}{\text{Total\_Rows\_in\_Frame}} * S$$



# Recovering ECDSA Keys

Calculating the ECDSA signing time

1. We extracted the series of frames associated with each signature from the video.
2. We filtered any series that started/ended during the transition time.
3. We calculated  $T_1$  = the execution time of all of the frames ( $F_{start+1} \dots F_{end+}$ ).
4. We calculated  $T_2$  = the execution time of the first frame in which the color changed ( $F_{start}$ ).
5. We calculated  $T_3$  = the execution time of the last frame in which the color changed ( $F_{end}$ ).
6. Sign time =  $T_1 + T_2 + T_3$ .



# Recovering ECDSA Keys

## Running the Minerva script

```
[ ] Guess: 0xa2aac4136847b714926f22d62bdd8fdf79ef73ce4d4e392763ef87c9bb2db4dc [3s]
[ ] Guess: 0x5d553beb97b848ec6d90dd29d422702042f786df59c9655d8fca42f941357075 [3s]
[ ] Guess: 0x2e30a32b12ecc2cf75af7eee4ec2de358daf17d9da1adfc2c3ad0f2ae2f5216 [3s]
[ ] Guess: 0xd1cf5cd3ed133d318a508111b13d21ca2f37e2d3ccfcbef7c77ef9d04e33d33b [3s]
[ ] Guess: 0x74eb1f28ecd016dbbe93cf5e19bb5cbce6a38f3759aa784a49bec30de697b8af [3s]
[ ] Guess: 0x8b14e0d6132fe925416c30a1e644a342d6436b764d6d263aa9fb07b515cb6ca2 [3s]
[ ] Guess: 0xc74ff93dc05dd634aeebe5d359e103dfac9e90c303c44b582c20dc86ae3410ca [3s]
[ ] Guess: 0x38b006c13fa229cc51141a2ca61efc20104869eaa353532cc798ee3c4e2f1487 [3s]
[ ] Guess: 0x549579732dd3b015f58dc1e20722a0163e6812aa97bfc800d417b6321f5908d [3s]
[ ] Guess: 0xab6a868bd22c4feb0a723e1df8dd5fe97e7ee8030f57d104e6784f5fda6d94c4 [3s]
[ ] Guess: 0xcc97a4db7d100dd1b0faed0f9ac7c0038053a96e4ef5117b501a74ce2270604c [3s]
[ ] Guess: 0x33685b2382eff22f4f0512f065383ffc3c93513f58228d09a39f55f4d9f2c505 [3s]
[ ] Guess: 0x2f22a4179c90a52374198d98f79830129dd7974adb839eedac49a618bf0cc142 [3s]
[ ] Guess: 0xd0dd5be7636f5add8be672670867cfed1f0f6362cb93ff97477024aa3d56640f [3s]
[ ] Guess: 0xe1fac964e8f6c10ef2a6c3bbb329e14d2ef0b32ed79825916877518fbe5bb4b4 [3s]
[ ] Guess: 0x1e05369a17093ef20d593c444cd61eb28df6477ecf7f78f38b4279333e07709d [3s]
[ ] Guess: 0xa1cb56b6eedb98ecbe5e3865be51010a9ff4bcf14feab71607fe117e3ebcf6b8 [3s]
[ ] Guess: 0x5e34a9481124671441a1c79a41aefef51cf23dbc572ce76eebbb944bda62e99 [3s]
[ ] Guess: 0x7ef4a6332959c6d93390459dd15b263bcae6ba04a1c8c44037cc1f2b2930d04f [3s]
[ ] Guess: 0x010b59cbd6a63927cc6fba622ea4d9c3f20040a9054eda44bbbedab97d3325502 [3s]
[ ] *** FOUND PRIVATE KEY *** : 0x010b59cbd6a63927cc6fba622ea4d9c3f20040a9054eda44bbbedab97d3325502 [3s]
[*] Finished attack.
```

We recovered a 256-bit ECDSA key from video footage obtained by an Internet-connected video camera located 16 meters away from the smartcard.

# Recovering ECDSA Keys



We recovered ECDSA keys from five other smartcard readers that we purchased on Amazon but from shorter distances (5-10 meters).

# Recovering ECDSA Keys



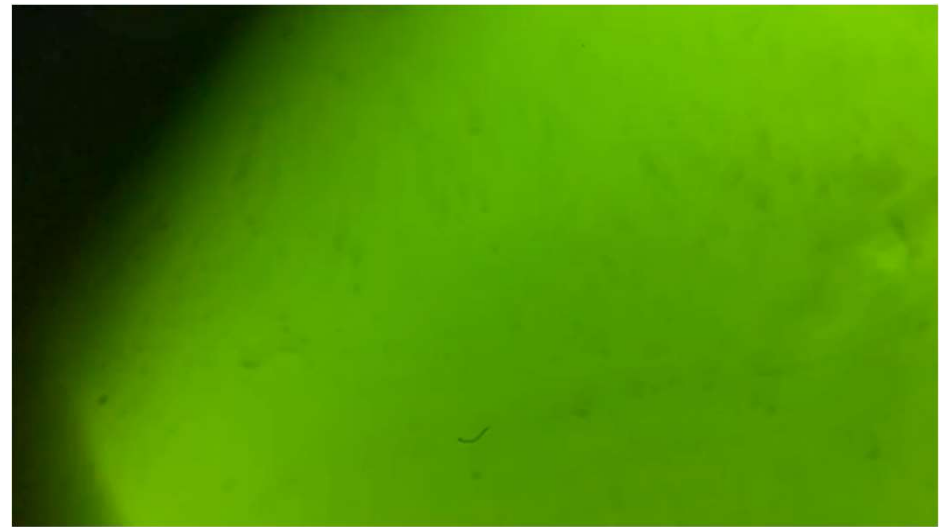
Now let's discuss recovering a SIKE key

# Recovering SIKE Keys

## HertzBleed Attack (2022)

1. The **Herzbleed attack** introduced a timing attack to recover a 378-bit SIKE key.
2. The Herzbleed attack exploited the fact that devices' dynamic voltage frequency stabilizers (DVFSs) have different execution times for different data.
3. For each index, attackers can craft a dedicated cryptogram, using the recovered bits (adaptive attack), and determine whether the value of the bit is similar or different from the previous bit based on a timing threshold.
4. The original attack was implemented using timing measurements obtained by querying the API of a server over the Internet ( $\text{Time}_{\text{response}} - \text{Time}_{\text{request}}$ ).
5. Due to the noise added by the network latency, it took a few days to recover the key.

# Recovering SIKE Keys



Objective: Recovering a 378-bit SIKE key from a Samsung Galaxy S8 using video footage of the power LED of Logitech Z120 USB speakers that were connected to the same USB hub used to charge the Galaxy S8 via an iPhone 13 Pro Max.



# Recovering SIKE Keys



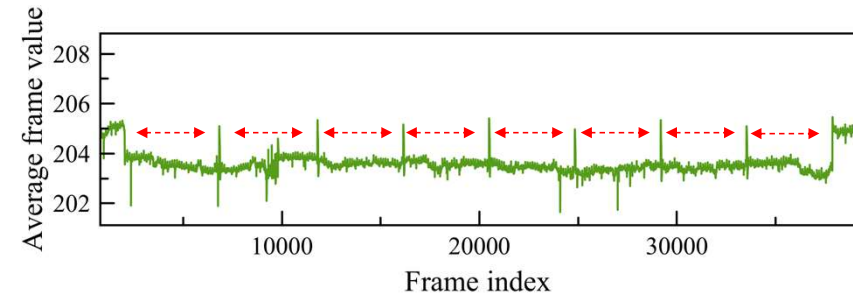
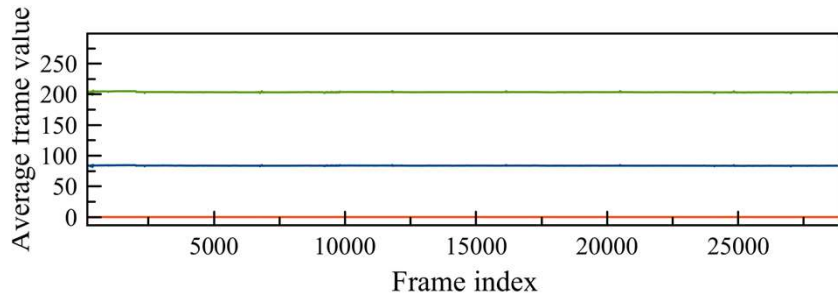
Idle



SIKE operations

The differences in the color of the power LED when the Galaxy S8 in idle and performing CPU operations cannot be detected by the naked eye.

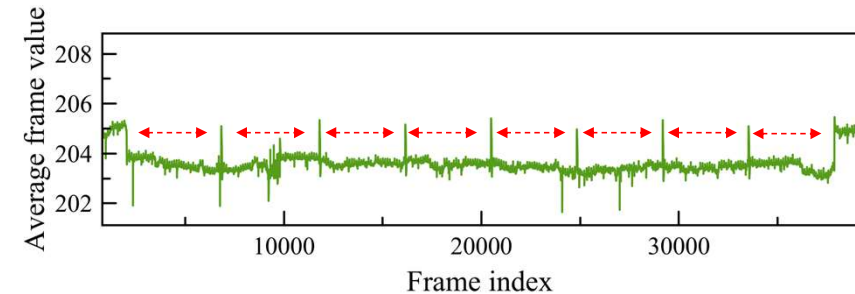
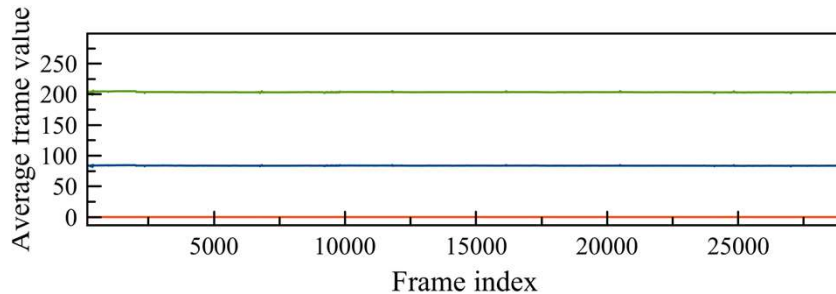
# Recovering SIKE Keys



**Eight consecutive iterations of SIKE operations (each iteration consisted of 100 SIKE operations)**

By averaging each frame into a single value and arranging the values in a time series, we can detect the beginning and end of iterations (based on the green channel).

# Recovering SIKE Keys

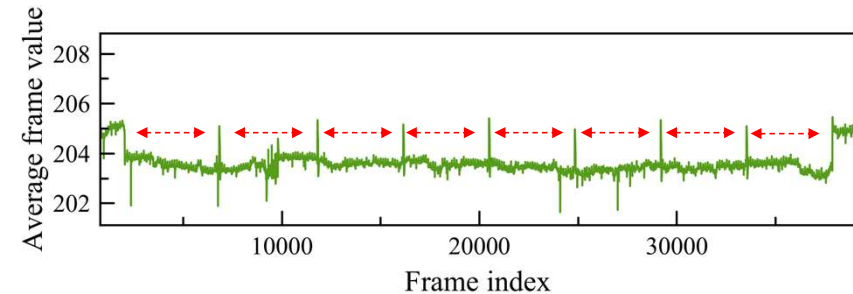
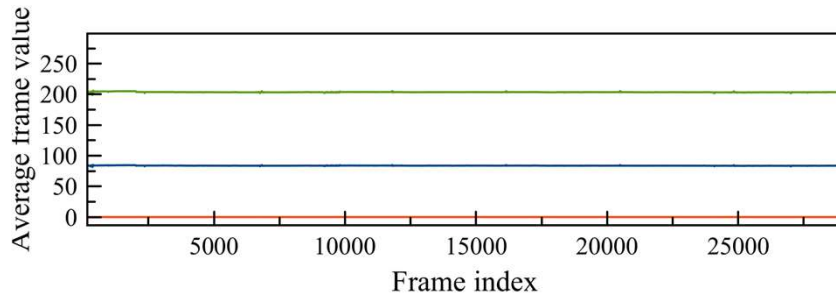


**Eight consecutive iterations of SIKE operations (each iteration consisted of 100 SIKE operations)**

By averaging each frame into a single value and arranging the values in a time series, we can detect the beginning and end of iterations (based on the green channel).

We can then calculate the iteration time based on the number of values between the peaks.

# Recovering SIKE Keys



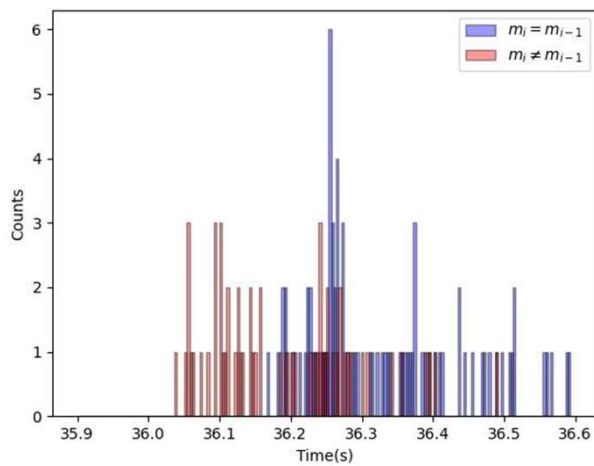
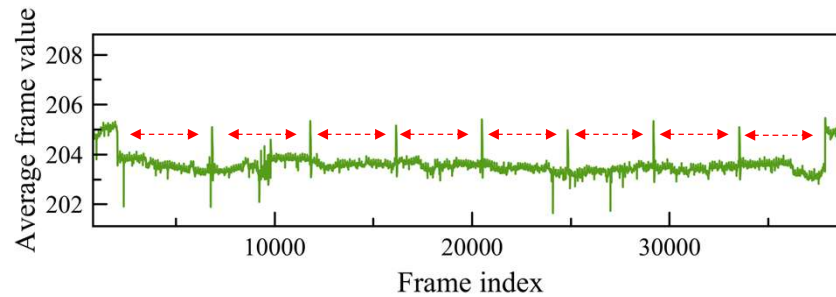
**Eight consecutive iterations of SIKE operations (each iteration consisted of 100 SIKE operations)**

By averaging each frame into a single value and arranging the values in a time series, we can detect the beginning and end of iterations (based on the green channel).

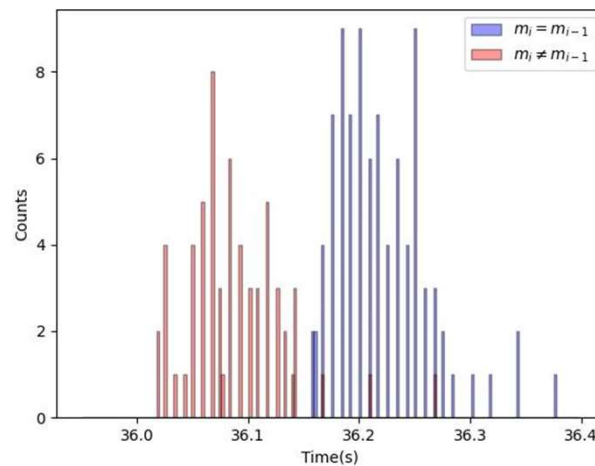
We can then calculate the iteration time based on the number of values between the peaks.

How accurate is the calculation?

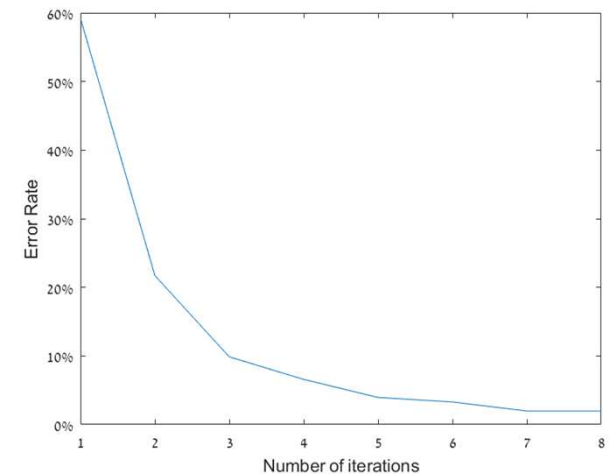
# Recovering SIKE Keys



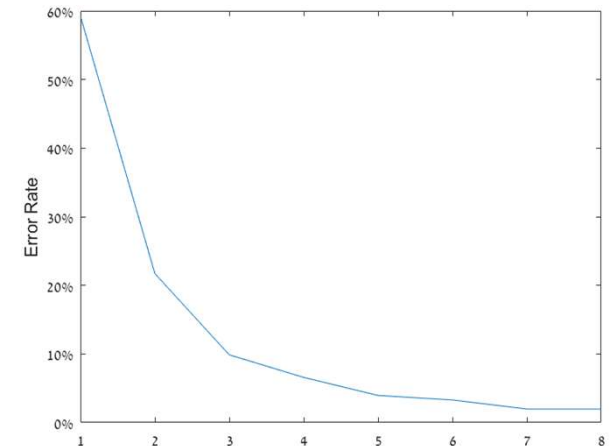
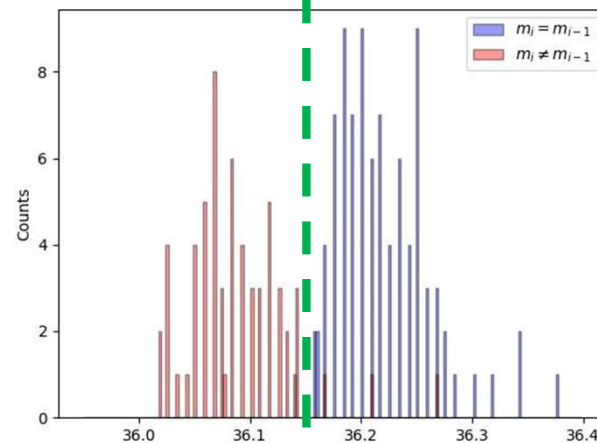
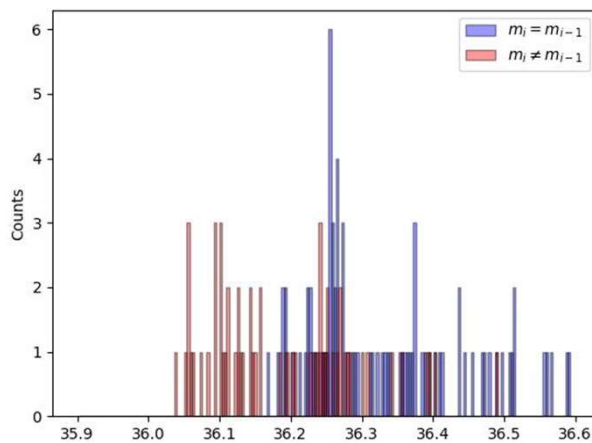
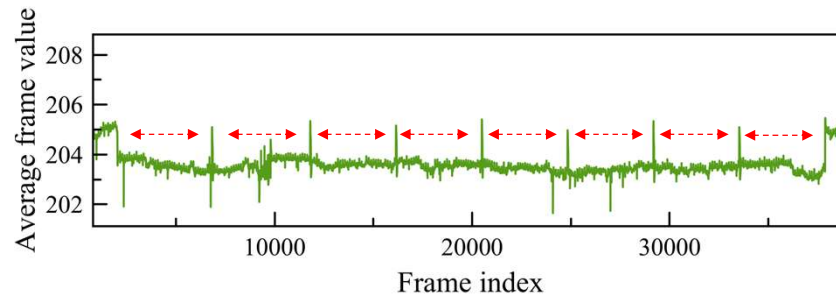
Distribution of the execution time based on the first iteration



Distribution of the execution time based on the minimum execution time of the eight iterations

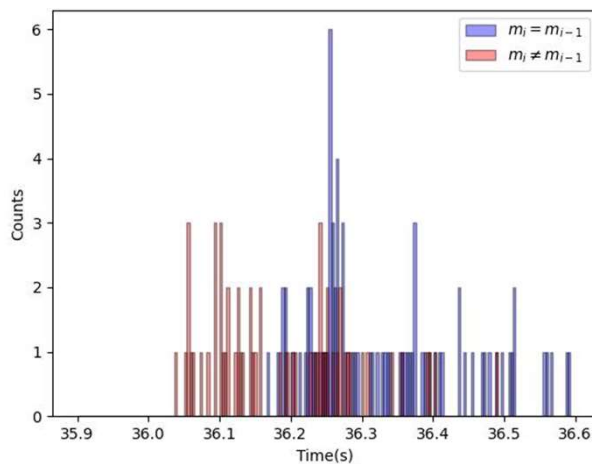
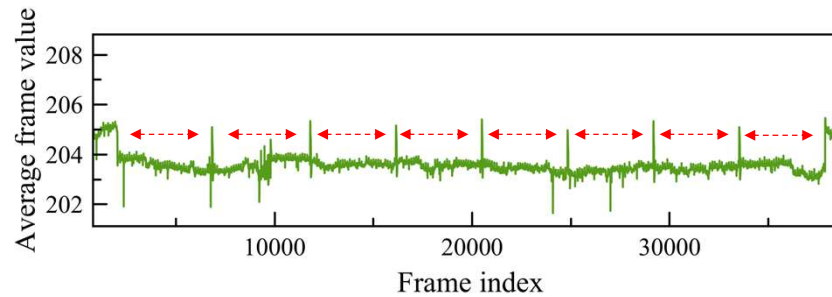


# Recovering SIKE Keys

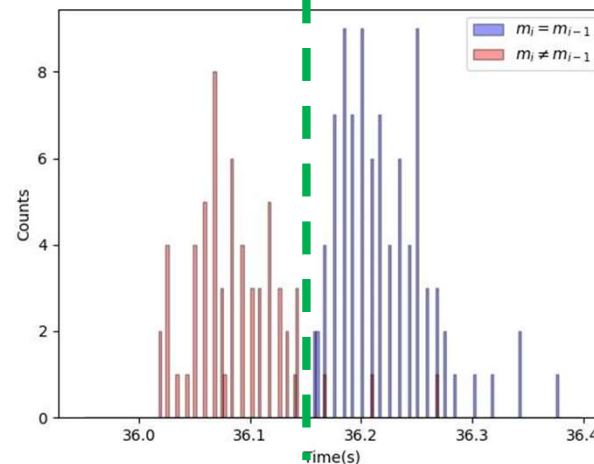


By calculating the minimum time of the eight iterations (per index) and setting a threshold of 36.15 s to distinguish between the two possible bits of the key's index, we could detect the bit with 99% accuracy.

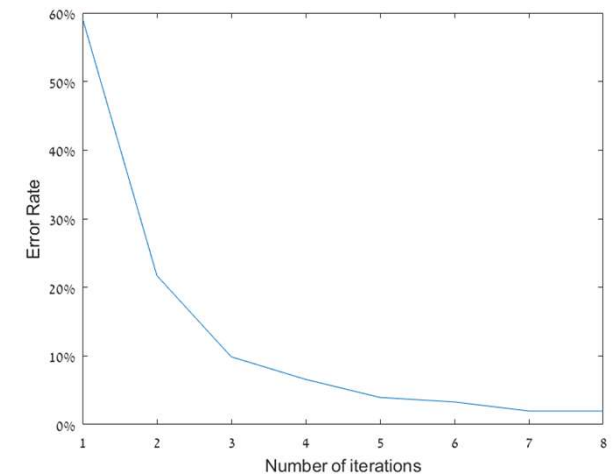
# Recovering SIKE Keys



Distribution of the execution time based on the first iteration

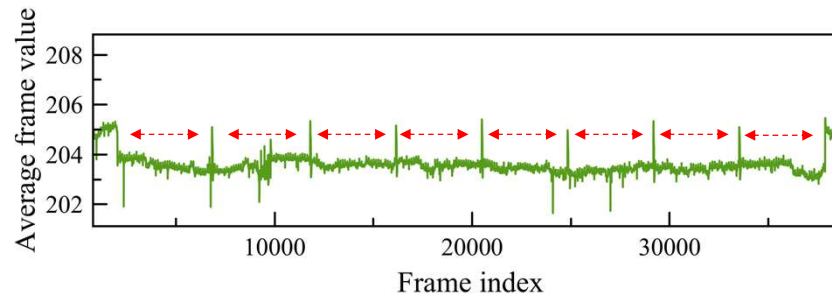


Distribution of the execution time based on the minimum execution time of the eight iterations

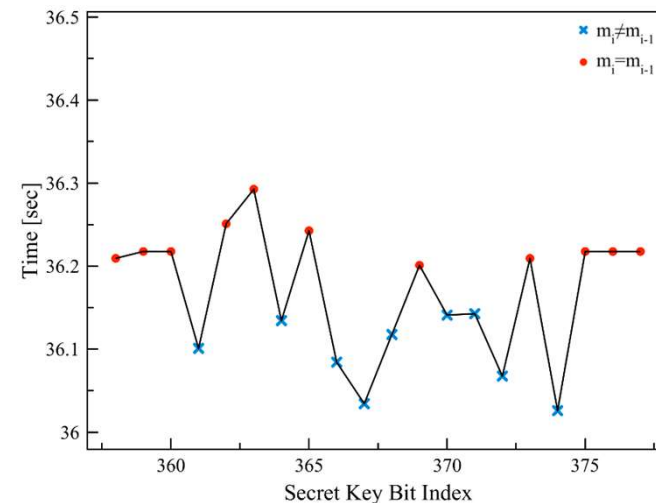
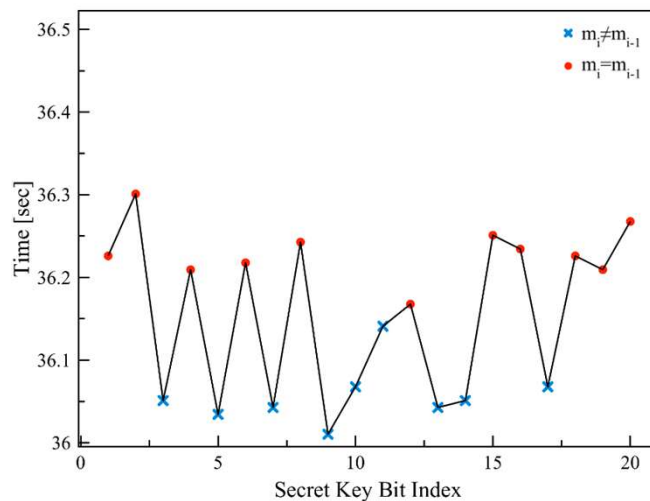


An error detecting mechanism will be needed to detect errors in the key's predictions.

# Recovering SIKE Keys



## Recovery of Key Bits



We recovered the entire 378-bit SIKE key from the Samsung Galaxy S8 with the use of error detecting and correcting mechanism.

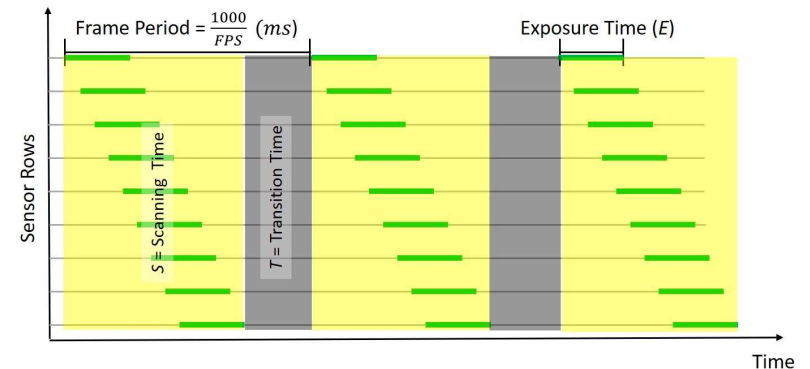


# Limitations

❖ The Attack Targets Weak IoT Devices (up to a level of a smartphone)



❖ Semi uniform distribution of the sampling



# **Takeaways**

1. Power LEDs are much more informative than one might imagine. In some cases, power LEDs leak confidential information that can be used to breach a user's privacy and information confidentiality.

# Takeaways

2. The potential of the attack is actually greater than shown in this work.
  - Some professional cameras support a much **higher rolling shutter rate** ( $\frac{1}{180,000}$ ) than the iPhone ( $\frac{1}{60,000}$ ).
  - Some professional cameras support a **higher bit depth** (10 and 12 bits) for the RGB space than the bit depth we used (8 bits).
  - Some professional cameras support **enhanced zoom capabilities** (x35) than the camera we used (x25).

# Takeaways

3. We expect that more and more devices will be exposed to video-based cryptanalysis each year.
  - Video cameras' specs are continuously improving (following Moore's Law).

|                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
|----------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                      |   |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| iPhone Model         |  |  |  |  |  |  |  |  |  |
| Year of Announcement | 2007                                                                                  | 2008                                                                                  | 2009                                                                                  | 2010                                                                                  | 2011                                                                                  | 2012                                                                                  | 2013                                                                                  | 2014                                                                                  | 2015                                                                                  |
| FPS&<br>Resolution   | No<br>Video                                                                           | No<br>Video                                                                           | 30fps<br>@480p                                                                        | 30fps<br>@720p                                                                        | 30fps@<br>1080p                                                                       | 30fps@<br>1080p                                                                       | 120fps<br>@720p                                                                       | 240fps<br>@720p                                                                       | 240fps<br>@720p                                                                       |

# Takeaways

3. We expect that more and more devices will be exposed to video-based cryptanalysis each year.
  - Video cameras' specs improve every year (following Moore's Law).
  - The number of functional IoT devices with limited CPUs (sensors, smart assistants, smartcards, robotic vacuum cleaners, smartphones, and video streamers) deployed continues to grow.

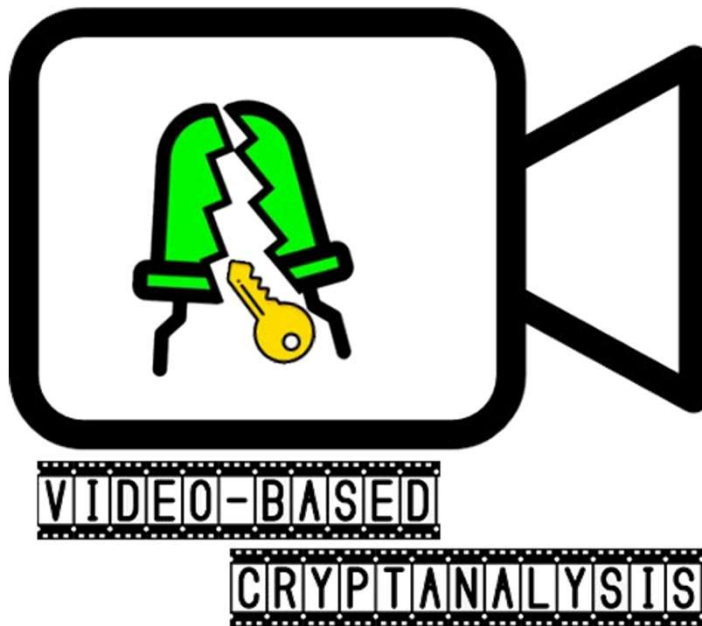
# Takeaways

4. It might be easier to compromise a target device with a malware and exfiltrate the key over the Internet.

However, video-based cryptanalysis is intended to extract keys from non-compromised devices using popular equipment.

# Q&A

- Thank you very much for attending this talk.



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Twitter: @ben\_nassi